

Lecture 8: Quantum mechanics and atomic structure 2

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

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Major points of last lecture

- **Quantum mechanics and atomic structure**
 1. **The hydrogen atom**
 2. **Shell model for many-electron atoms**
 3. **Aufbau principle and electron configuration**
 4. **Shells and the periodic table: photoelectron spectroscopy**
 5. **Periodic properties and electronic structure**

The hydrogen atom: Schrödinger equation

the Schrödinger equation (time-independent)

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V \right] \psi = E\psi$$

动能算符 势能算符

time independent Schrödinger equation for H atom

$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

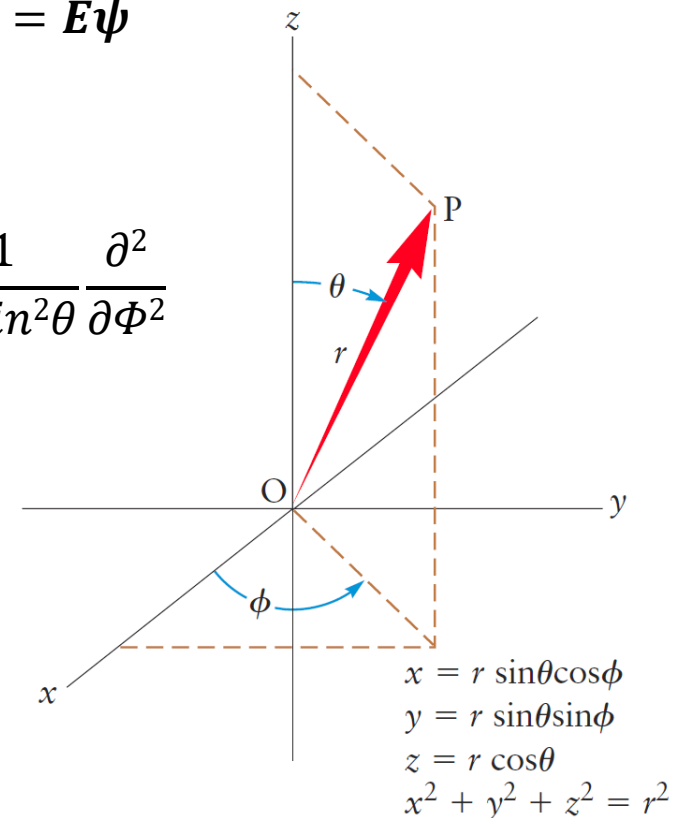
Cartesian coordinates (笛卡尔坐标)

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \sin\theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2}{\partial \phi^2}$$

Spherical coordinates (球坐标)

Solve the Schrödinger equation

$$\psi_{nlm}(r, \theta, \phi)$$



The hydrogen atom: wave functions

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

radial part angular part

For each quantum state (n, ℓ, m), solution of the Schrödinger equation provides a wave function:

TABLE 5.2 Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$

Radial Part $R_{n\ell}(r)$

$$\ell = 0 \left\{ Y_s = \left(\frac{1}{4\pi} \right)^{1/2}$$

$$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$$

$$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$$

$$\ell = 1 \left\{ \begin{array}{l} Y_{p_x} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi \\ Y_{p_y} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \sin \phi \\ Y_{p_z} = \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta \end{array} \right.$$

$$R_{2p} = \frac{1}{2\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$$

$$\ell = 2 \left\{ \begin{array}{l} Y_{d_{z^2}} = \left(\frac{5}{16\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \\ Y_{d_{xz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \cos \phi \\ Y_{d_{yz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \sin \phi \\ Y_{d_{xy}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \sin 2\phi \\ Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \cos 2\phi \end{array} \right.$$

$$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0}$$

$$a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

The hydrogen atom: energy levels

$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$$

Solutions of the Schrödinger equation for the one-electron atom exist only for **particular values of the energy**

$$E = E_n = -\frac{Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} \quad n = 1, 2, 3, \dots$$

$$E_n = -\frac{Z^2}{n^2} \quad (\text{rydberg}) \quad n = 1, 2, 3, \dots$$

The integer **n** (**principal quantum number**, 主量子数) indexes the individual energy levels. These are identical to the energy levels predicted by the Bohr theory.

Quantization arises because of the physical boundary conditions imposed on the solutions to the Schrodinger equation (需要满足边界条件) rather than from making arbitrary assumptions about the angular momentum.

The hydrogen atom: quantum numbers

The Schrödinger equation **also quantizes L^2** , the square magnitude of the angular momentum (L), as well as **L_z** , the projection of the angular momentum along the z-axis (L_z).

$$L^2 = \ell(\ell + 1) \frac{h^2}{4\pi^2} \quad \ell = 0, 1, \dots, n - 1$$

The **angular momentum quantum number ℓ (角量子数)**, may take on any integral value from 0 to $n-1$

$$L_z = m \frac{h}{2\pi} \quad m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

The **magnetic quantum number m (磁量子数)**, may take on any integral value from $-\ell$ to ℓ

Its value governs the behavior of the atom in an external magnetic field.

The hydrogen atom: quantum numbers

$$n = 1, 2, 3, \dots$$

$$\ell = 0, 1, \dots, n - 1$$

$$m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

When $n=1$. Only one set of quantum states: $n=1, \ell=0, m=0$

When $n=2$. Four sets of quantum states: $(2, 0, 0); (2, 1, -1); (2, 1, 0); (2, 1, 1)$

When $n=n$. $1+3+5+\dots+(2n-1)=n^2$ sets of quantum states

- Each set **(n, ℓ, m)** identifies a specific **quantum state** of the atom in which the electron has energy equal to E_n .
- When $n > 1$, a total of n^2 specific quantum states correspond to the single energy level E_n ; consequently, this set of states is said to be **degenerate (简并)**.

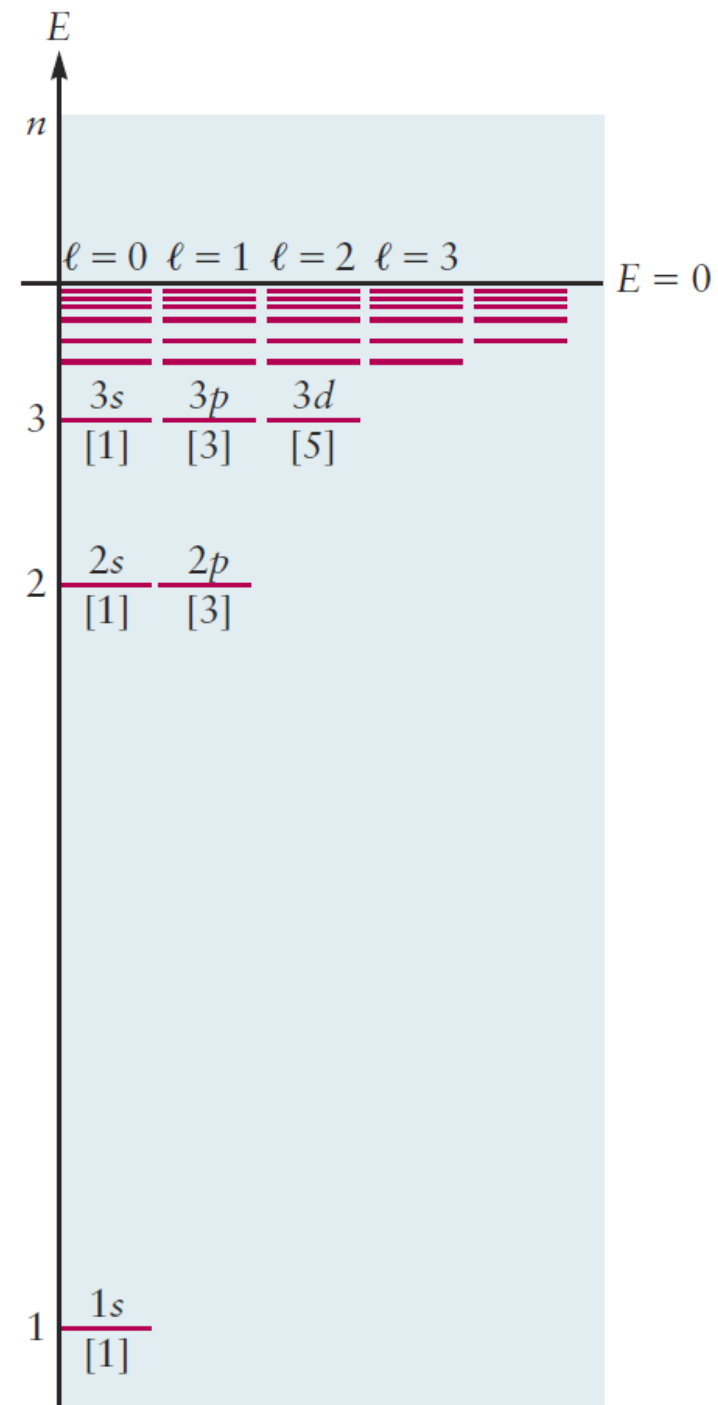
The hydrogen atom: quantum numbers

It is conventional to label specific states by replacing the angular momentum quantum number with a letter

$\ell = 0$ with **s**, $\ell = 1$ with **p**, $\ell = 2$ with **d**, $\ell = 3$ with **f**, $\ell = 4$ with **g**

TABLE 5.1 Allowed Values of Quantum Numbers for One-Electron Atoms

n	1	2	3
ℓ	0	0, 1	0, 1, 2
m	0	0, -1 , $+1$	0, -1 , $+1$, -2 , $+2$
Number of degenerate states for each ℓ	1	1, 3	1, 3, 5
Number of degenerate states for each n	1	4	9



The hydrogen atom: orbital

A wave function $\psi_{n\ell m}(r, \theta, \phi)$ for a one-electron atom in the state (n, ℓ, m) is called an **orbital**.

When the one-electron atom is in state (n, ℓ, m) , it is conventional to say the electron is “in an (n, ℓ, m) orbital.”

When an electron has energy, total angular momentum, and z component of angular momentum values corresponding to the quantum numbers n, ℓ, m , the probability density of finding the electron at the point (r, θ, ϕ) is given by $\psi_{n\ell m}^2(r, \theta, \phi)$.

Orbital **is not** some sort of “region in space” inside which the electron is confined.

vs. Bohr model for H atom

The hydrogen atom: s orbitals

s ORBITALS: corresponding to $\psi_{n\ell m}(r, \theta, \phi)$ with $\ell=0$ ($m = 0$ as well).

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$$

$$\ell = 0 \left\{ Y_s = \left(\frac{1}{4\pi} \right)^{1/2}$$

$$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$$

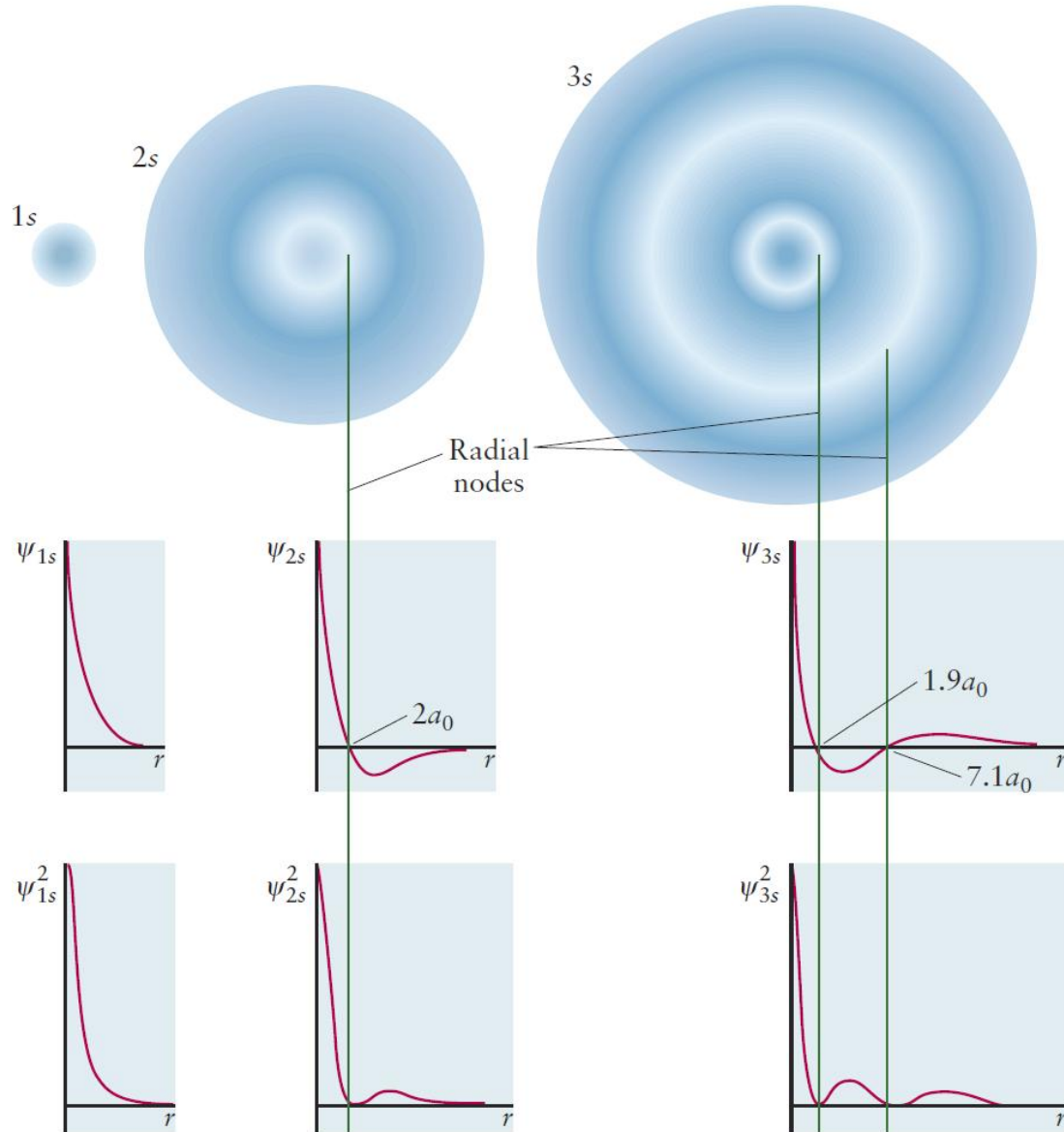
$$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0} \quad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

$$\psi_{1s} = \frac{1}{\sqrt{\pi}} a_0^{-3/2} \cdot \exp\left(-\frac{r}{a_0}\right)$$

$$\psi_{2s} = \frac{1}{4\sqrt{2\pi}} a_0^{-3/2} \cdot \left(2 - \frac{r}{a_0}\right) \exp\left(-\frac{r}{2a_0}\right)$$

The hydrogen atom: s orbitals



$$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$$

$$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$$

$$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$$

The hydrogen atom: p orbitals

p ORBITALS: $\ell=1$, $m=-1, 0, 1$

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$$

$$\ell = 1 \left\{ \begin{array}{l} Y_{p_x} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi \\ Y_{p_y} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \sin \phi \\ Y_{p_z} = \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta \end{array} \right.$$

$$R_{2p} = \frac{1}{2\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$$

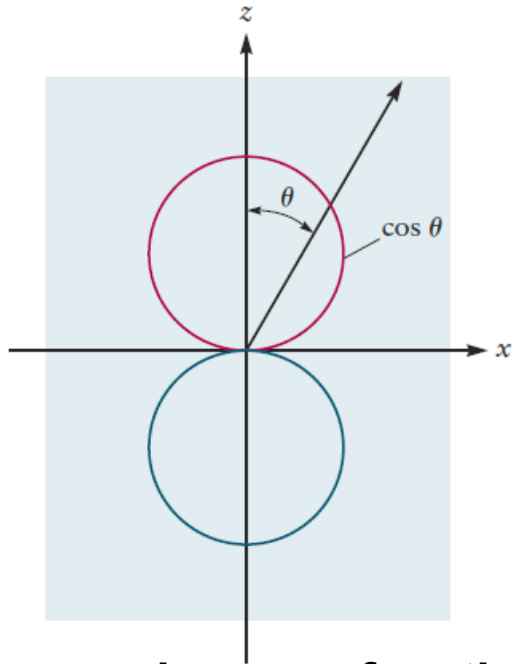
$$\sigma = \frac{Zr}{a_0} \quad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

Orbitals with angular quantum numbers
different from 0 are not spherically symmetric

The hydrogen atom: p orbitals

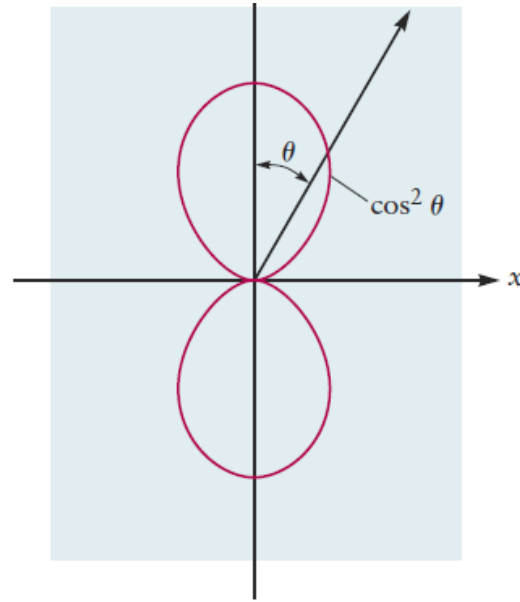
Angular wave function

1 angular nodes



The angular wave function for the p_z orbital. The p_x and p_y orbitals are oriented along the x - and y -axis, respectively.

$$\ell = 1 \begin{cases} Y_{p_x} = \left(\frac{3}{4\pi}\right)^{1/2} \sin \theta \cos \phi \\ Y_{p_y} = \left(\frac{3}{4\pi}\right)^{1/2} \sin \theta \sin \phi \\ Y_{p_z} = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta \end{cases}$$



The square of the angular wave function for the p_z orbital.

Radial wave function

$n - \ell - 1$ radial nodes

$$R_{2p} = \frac{1}{2\sqrt{6}} \left(\frac{Z}{a_0}\right)^{3/2} \sigma \exp(-\sigma/2)$$

$$R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0}\right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$$

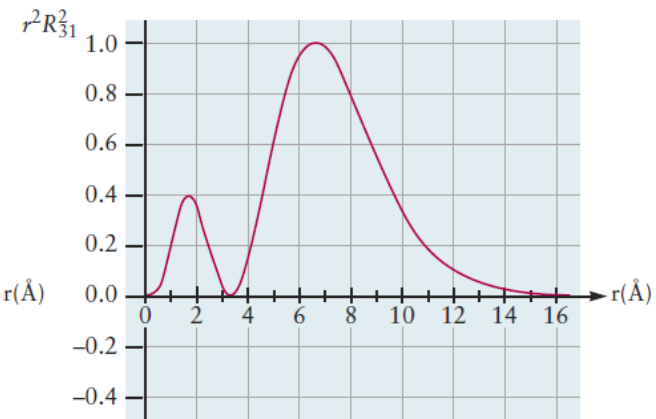
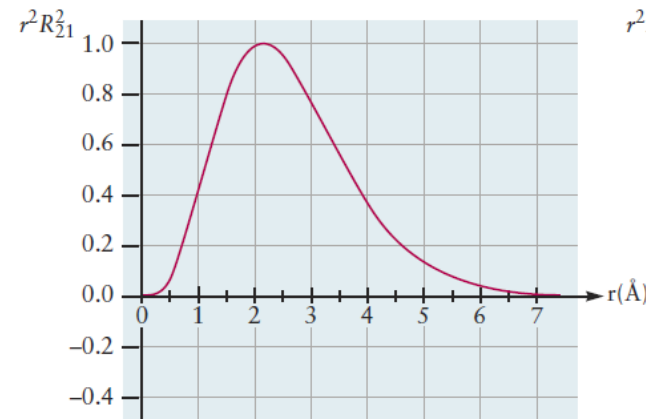
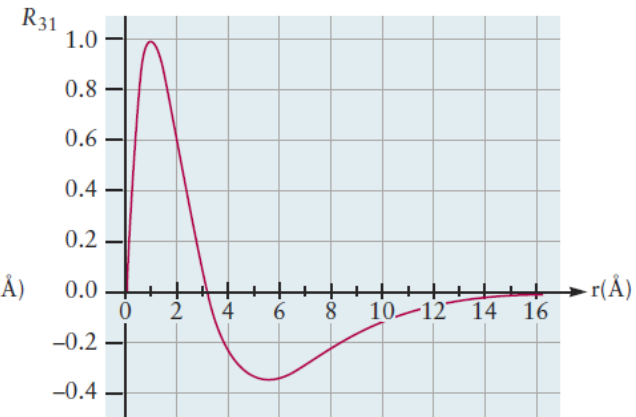
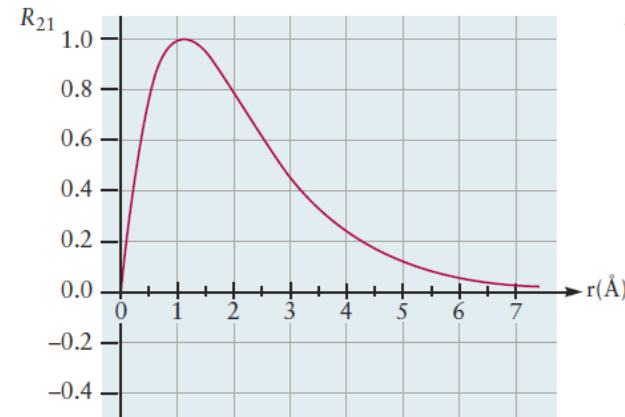


FIGURE 5.10 Radial wave functions $R_{n\ell}$ for np orbitals and the corresponding radial probability densities $r^2 R_{n\ell}^2$.

The hydrogen atom: d orbitals

d ORBITALS: $\ell=2$, $m=-2, -1, 0, 1, 2$

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi)$$

$$\ell = 2 \left\{ \begin{array}{l} Y_{d_{z^2}} = \left(\frac{5}{16\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \\ Y_{d_{xz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \cos \phi \\ Y_{d_{yz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \sin \phi \\ Y_{d_{xy}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \sin 2\phi \\ Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \cos 2\phi \end{array} \right.$$

$$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 \exp(-\sigma/3)$$

$$\sigma = \frac{Zr}{a_0}$$

$$a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$$

The hydrogen atom: d orbitals

d orbital: 2 angular nodes, n-3 radial nodes. n-1 total nodes

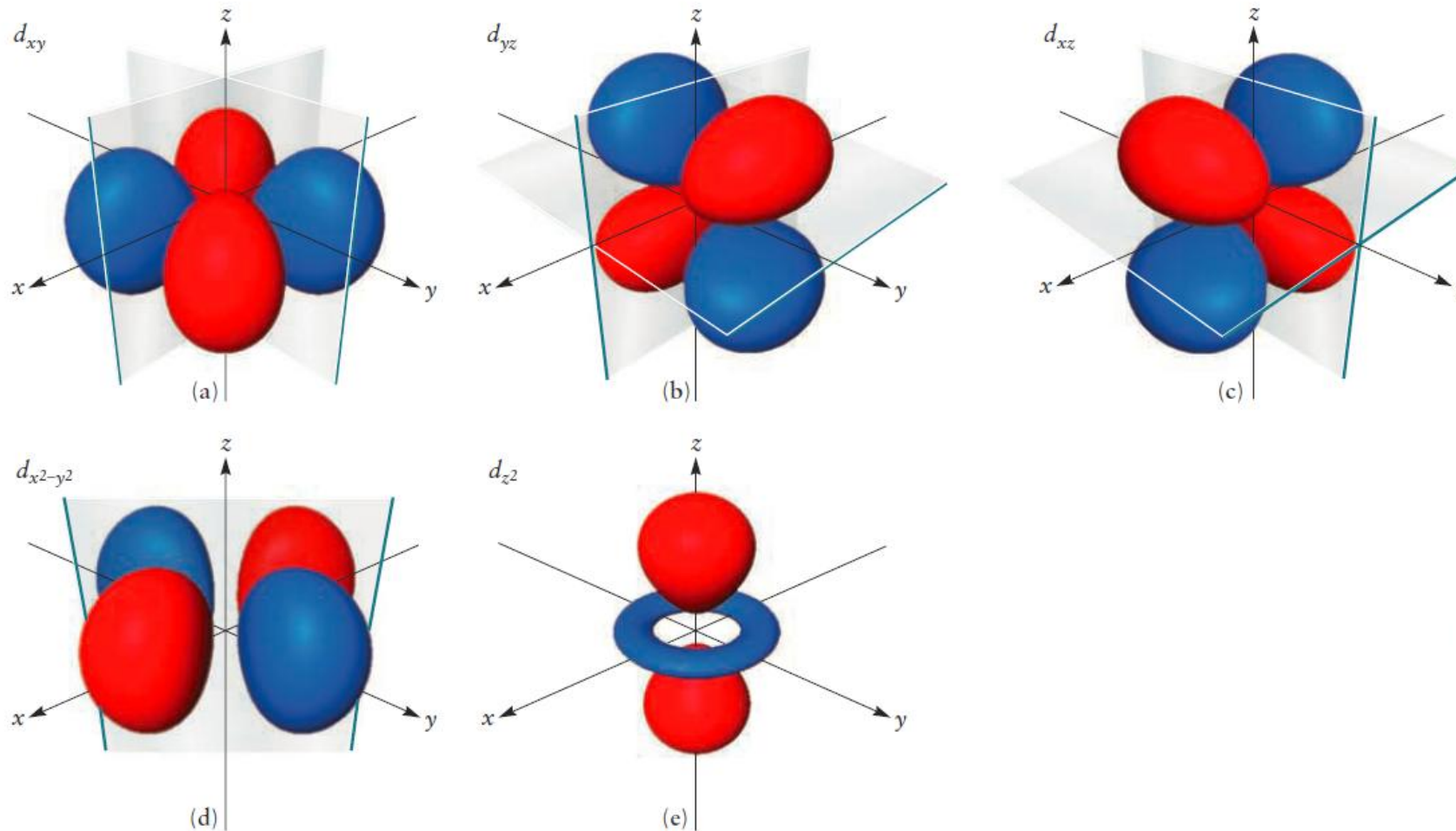
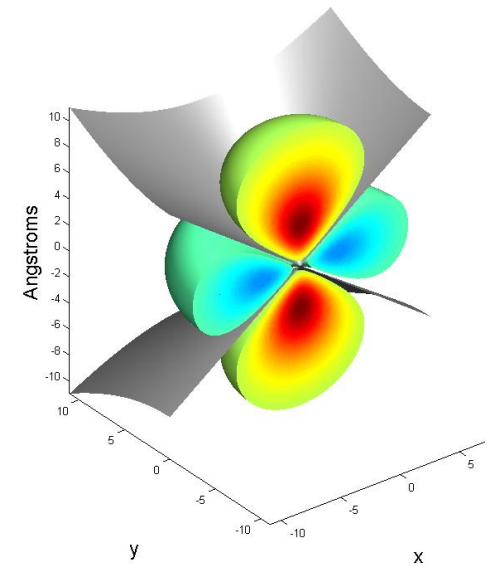


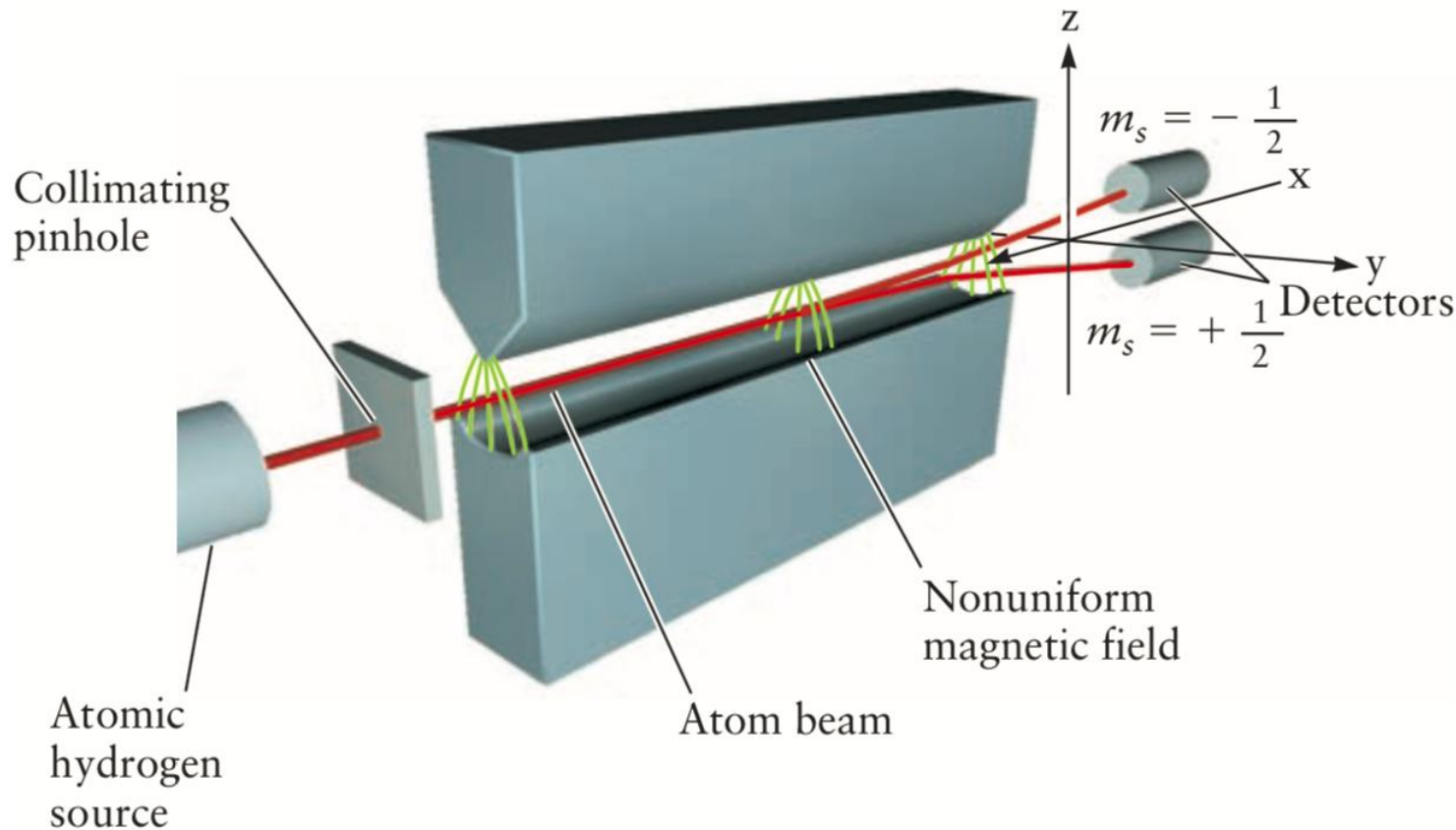
FIGURE 5.13 The shapes of the five 3d orbitals, with phases and nodal surfaces indicated. The "down-axis" view shows the shapes of the first four orbitals (a)–(d) when viewed down the appropriate axis; the specific example shown here is the d_{xy} orbital viewed down the z-axis.



The hydrogen atom: summary

- For a given ℓ : an **increase in n** leads to an **increase in the average distance of the electron from the nucleus**, and the **size of the orbital**
- An orbital (quantum numbers n and ℓ): **ℓ angular nodes** and **$n - \ell - 1$ radial nodes**, giving a total of **$n-1$ nodes**. An **angular node is defined by a plane**. A **radial node is defined by a spherical surface**.
- As **r approaches 0**, **$\psi(r, \theta, \phi)$ vanishes for all orbitals except s orbitals**; thus, only an electron in an **s orbital** can “penetrate to the nucleus,” that is, **have a finite probability of being found right at the nucleus**.

The hydrogen atom: electron spin



That the original beam **is split into only two well-defined beams** in this experiment demonstrates the unexpected fact that **the orientation of the magnetic moment of the electron is quantized**.

The result of this experiment is explained by introducing a fourth quantum number, m_s , which can take on two values, conventionally chosen to be **+1/2 and -1/2**. For historical reasons, the fourth quantum number is referred to as the **spin quantum number** 自旋量子数.

The hydrogen atom: quantum numbers

$2n^2$ sets of quantum states

$$n = 1, 2, 3, \dots$$

主量子数

$$\ell = 0, 1, \dots, n - 1$$

角量子数

$$m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

磁量子数

$$m_s = +1/2 \text{ or } -1/2$$

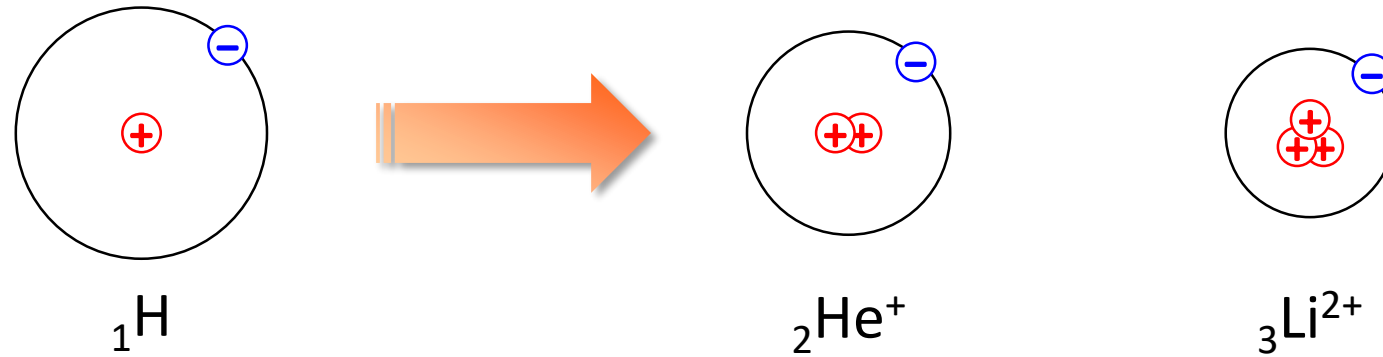
自旋量子数

$\ell = 0$ with s, $\ell = 1$ with p, $\ell = 2$ with d, $\ell = 3$ with f, $\ell = 4$ with g

Major points of today's lecture

- **Quantum mechanics and atomic structure**
 1. The hydrogen atom
 2. **Shell model for many-electron atoms**
 3. Aufbau principle and electron configuration
 4. Shells and the periodic table: photoelectron spectroscopy
 5. Periodic properties and electronic structure

Shell model for many-electron atoms: Hydrogen-Like Ions



Coulomb potential V

$$-\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$-\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

1s orbital $\psi_{1s} \propto \exp(-r / a_0)$

$$\exp(-Zr / a_0)$$

Orbital radius r_n

$$n^2 a_0$$

$$\frac{n^2}{Z} a_0$$

Orbital energy E_n

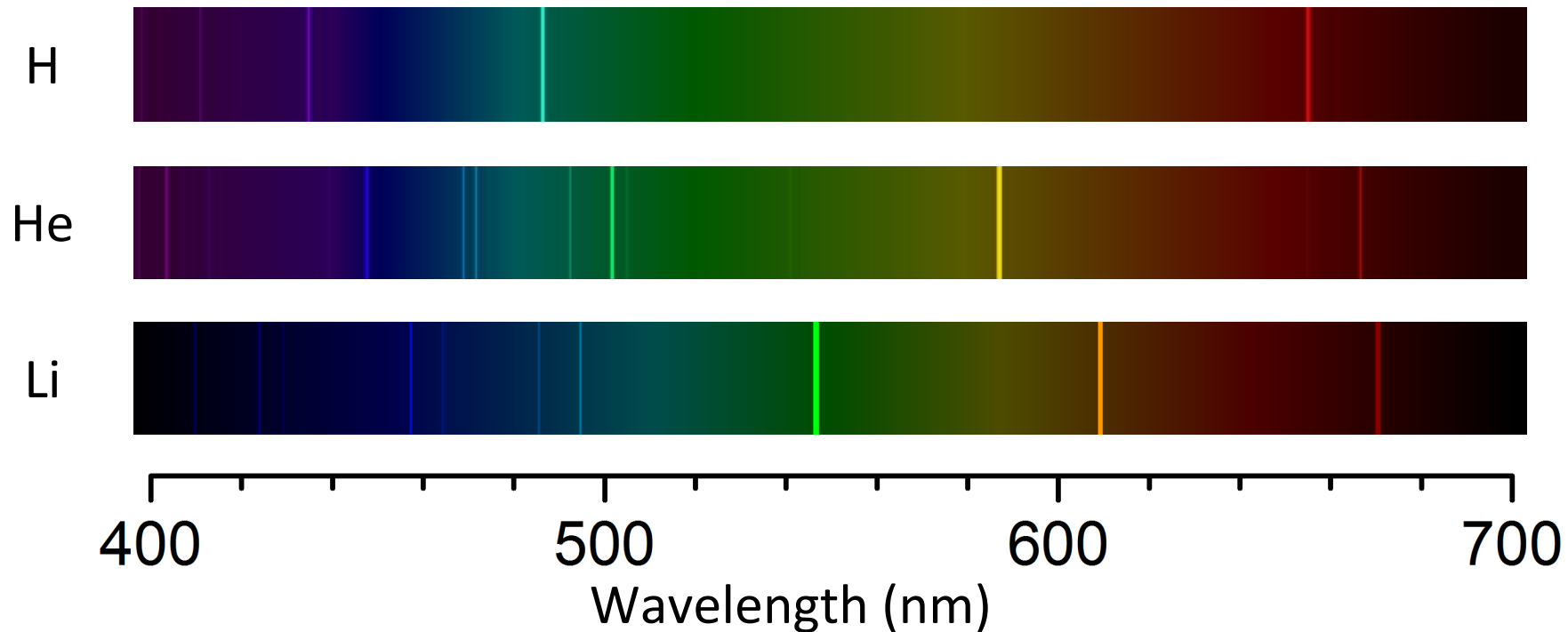
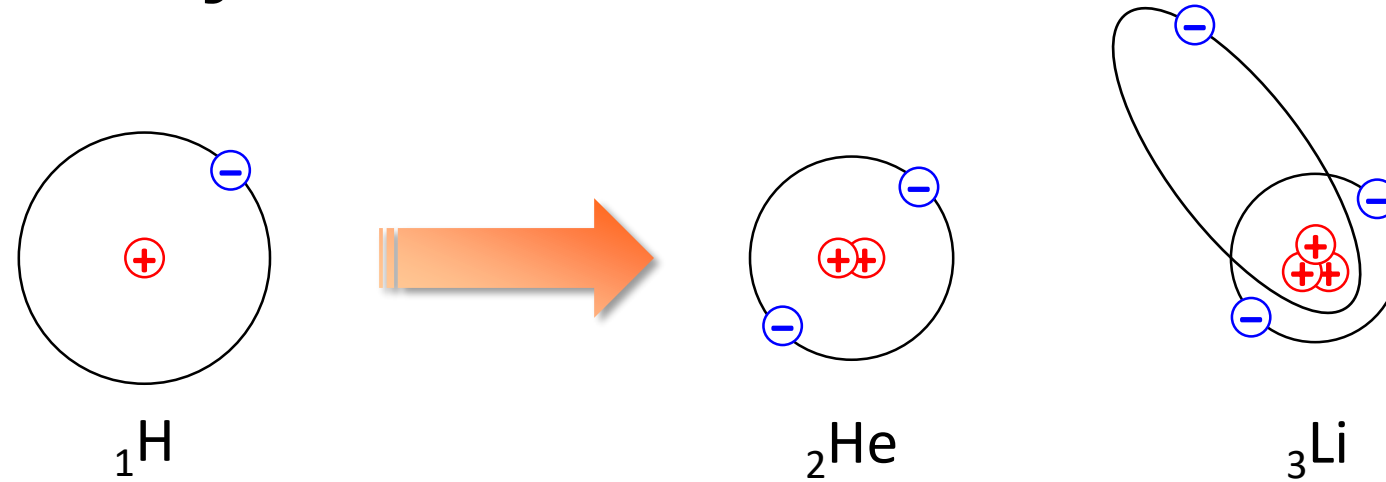
$$-\frac{\text{Ry}}{n^2}$$

$$-Z^2 \frac{\text{Ry}}{n^2}$$

$$a_0 = 53 \text{ pm}$$

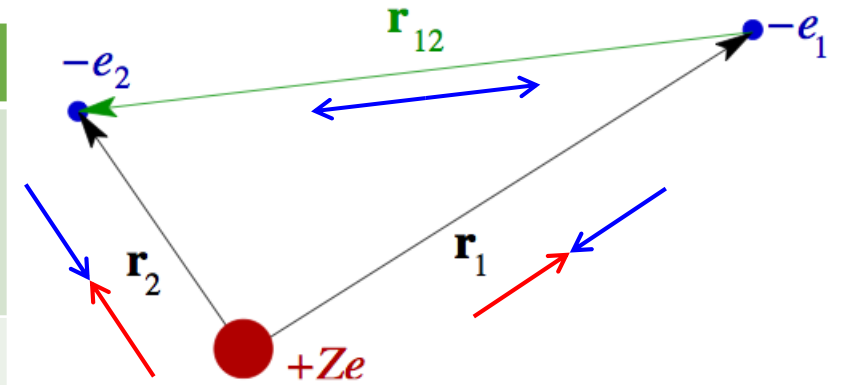
$$\text{Ry} = 13.6 \text{ eV}$$

Shell model for many-electron atoms: From H to He and Li



Shell model for many-electron atoms: Single- vs. Many-Electron Atoms

	H, He ⁺ , Li ²⁺	He, Li, ...
Interactions	Z-e attraction	Z-e attraction e-e repulsion electron spins
Orbitals	e in a single orbital	e in multiple orbitals
Coordinates	Z at O, e at r	Z at O, e_1 at r_1 , e_2 at r_2 , ...
Schrödinger Eqn.	Analytical solutions	Approximate solutions



An electron is attracted by $+Z$ but repelled by $-(Z-1)e$

$$\psi = \psi(x_1, y_1, z_1) \quad \left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r} \right] \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2) \quad \frac{-\hbar^2}{2m_e} \left[\left(\frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial y_1^2} + \frac{\partial^2 \psi}{\partial z_1^2} \right) + \left(\frac{\partial^2 \psi}{\partial x_2^2} + \frac{\partial^2 \psi}{\partial y_2^2} + \frac{\partial^2 \psi}{\partial z_2^2} \right) \right] + \left(-\frac{2e^2}{4\pi\epsilon_0 r_1} - \frac{2e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}} \right) \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$$

Shell model for many-electron atoms: Chemistry and Mathematics



Paul A. M. Dirac
(1902–1984)

The fundamental laws necessary for the mathematical treatment of a large part of physics and the whole of chemistry are thus **completely known**, and the difficulty lies only in the fact that application of these laws leads to equations that are **too complex to be solved**.

The many-electron Schrödinger Equation can only be solved
approximately.

Shell model for many-electron atoms: Hartree Orbitals

Self-consistent field (SCF) orbital approximation method (自洽场近似法):

1. generates **a set of approximate one-electron orbitals** (ψ_α), and associated energy levels (ϵ_α), reminiscent of those for the H atom.
2. The electronic structure of an atom with atomic number Z is then “built up” by placing Z electrons into these orbitals in accordance with certain rules

$$\frac{-\hbar^2}{2m_e} \left[\left(\frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial y_1^2} + \frac{\partial^2 \psi}{\partial z_1^2} \right) + \left(\frac{\partial^2 \psi}{\partial x_2^2} + \frac{\partial^2 \psi}{\partial y_2^2} + \frac{\partial^2 \psi}{\partial z_2^2} \right) \right] + \left(-\frac{2e^2}{4\pi\epsilon_0 r_1} - \frac{2e^2}{4\pi\epsilon_0 r_2} + \frac{e^2}{4\pi\epsilon_0 r_{12}} \right) \psi = E\psi$$

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2)$$

$$\varphi_\alpha(r_1)\varphi_\beta(r_2)$$

Shell model for many-electron atoms: Hartree Orbitals

Hartree's method: three assumptions

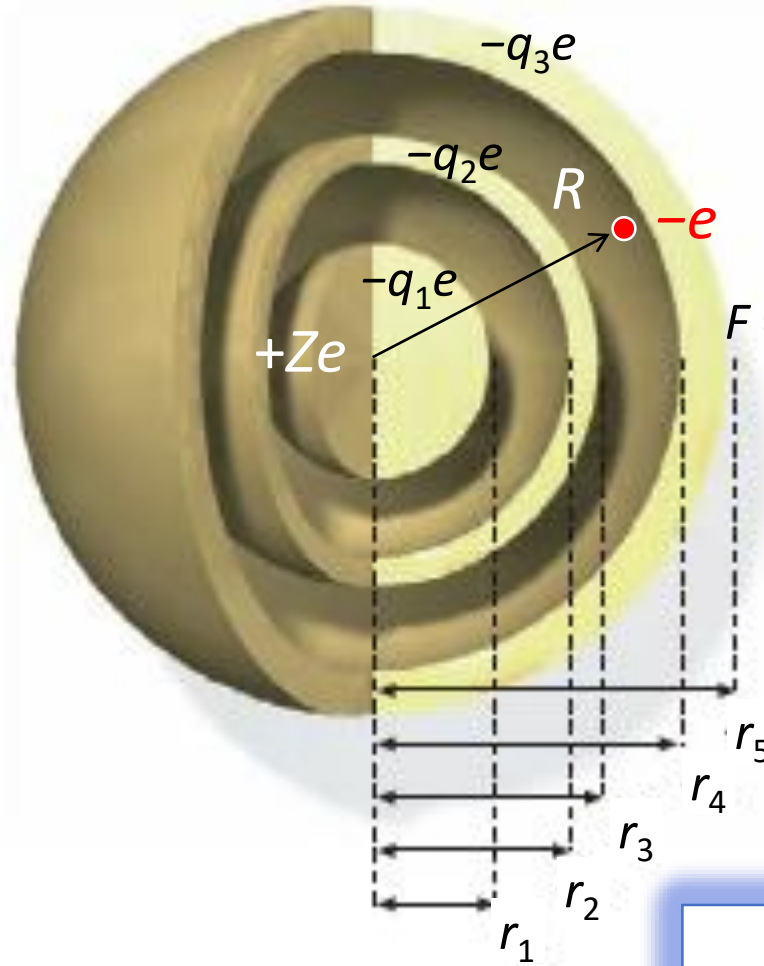
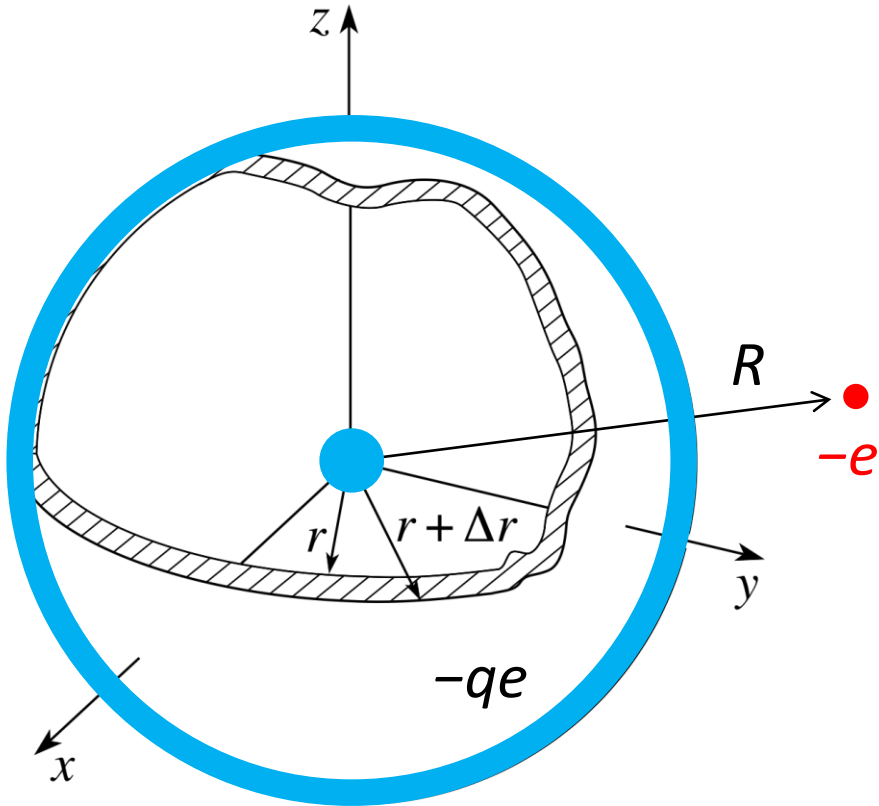
- Each electron moves in **an effective field** created by the nucleus and all the other electrons, and the effective field for electron i **depends only on its position** r_i .
- The effective field for electron i is obtained by averaging its Coulomb potential interactions with each of the other electrons over all the positions of the other electrons: **r_i is the only coordinate in the description.**
- The effective field is spherically symmetric: no angular dependence.

Li atom $\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$

The orbital approximation for atoms $\psi_{\text{atom}} = \varphi_{\alpha}(r_1)\varphi_{\beta}(r_2)\varphi_{\gamma}(r_3)$

The 2nd and 3rd assumptions in effect convert the exact Schrödinger equation for the atom into a set of simultaneous equations for the **unknown effective field and the unknown one-electron orbitals**. These equations must be **solved by iteration until a self-consistent solution** is obtained

Hartree Orbitals: effective nuclear charge



$$F = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{R^2} + \frac{1}{4\pi\epsilon_0} \frac{q_1e^2}{R^2} + \frac{1}{4\pi\epsilon_0} \frac{q_2e^2}{R^2}$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{(Z - q_1 - q_2)e^2}{R^2}$$

$$= -\frac{1}{4\pi\epsilon_0} \frac{Z_{\text{eff}}e^2}{R^2}$$

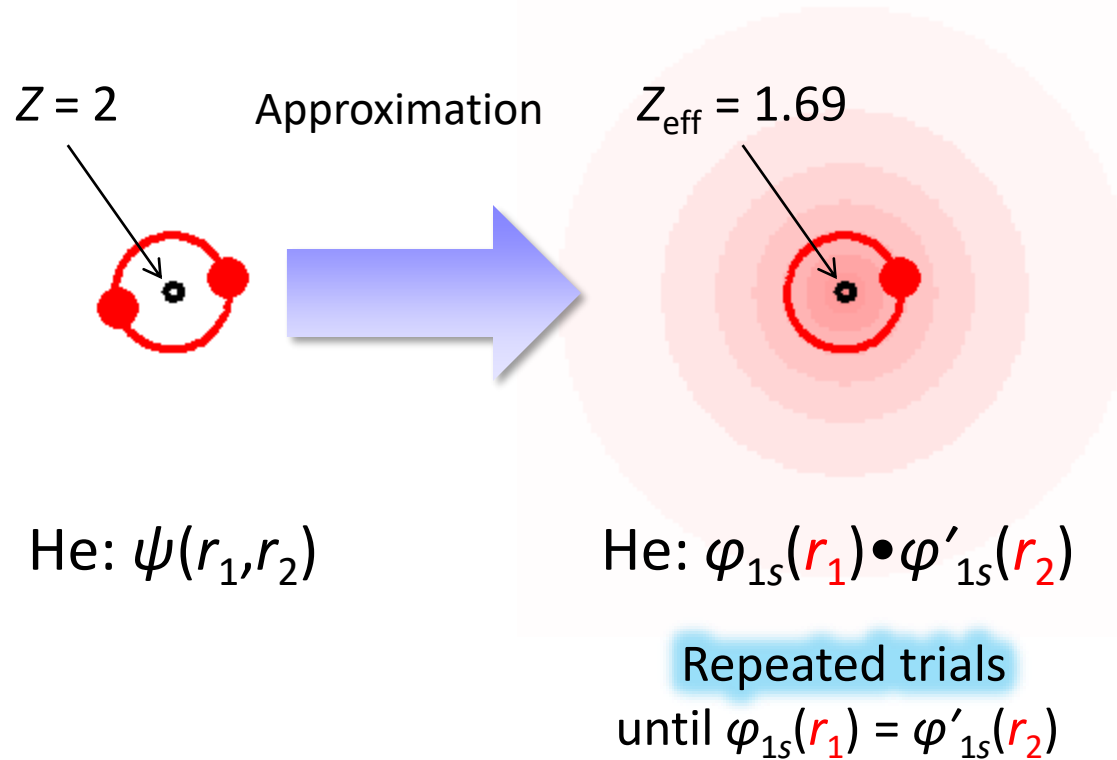
$$F = \frac{1}{4\pi\epsilon_0} \frac{qe^2}{R^2} \text{ when } R \geq r$$

$$F = 0 \text{ when } R < r$$

$$V = -\frac{1}{4\pi\epsilon_0} \frac{Z_{\text{eff}}e^2}{R}, \quad Z_{\text{eff}} = Z - q$$

Z_{eff} is **effective** nuclear charge.

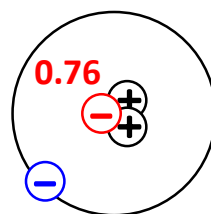
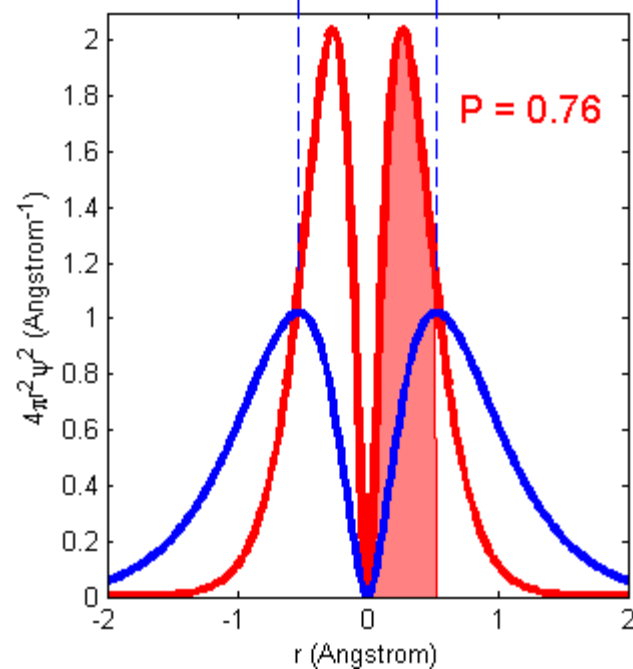
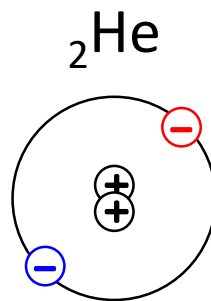
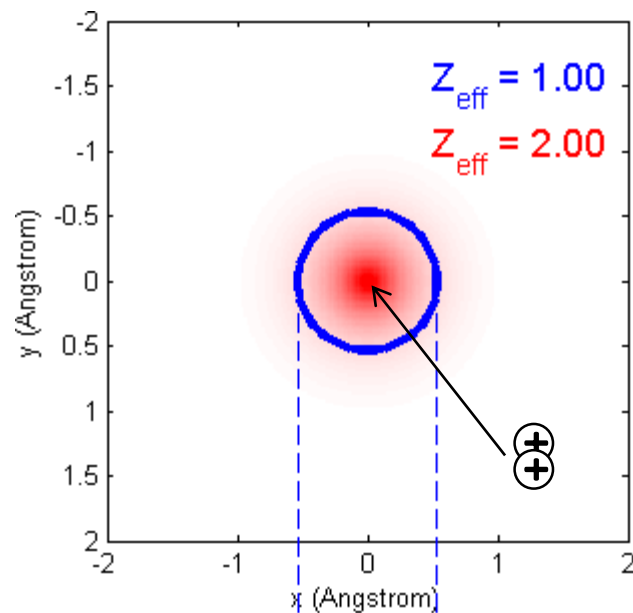
Hartree Orbitals: effective nuclear charge



Douglas Hartree
(1897–1958)

$$\varphi_{1s}(r) \propto \exp(-Z_{\text{eff}} r / a_0), \text{ } Z_{\text{eff}} \text{ is effective nuclear charge}$$

SCF Example: Helium



$$Z_{\text{eff}} = 2 - 0.76 = 1.24$$

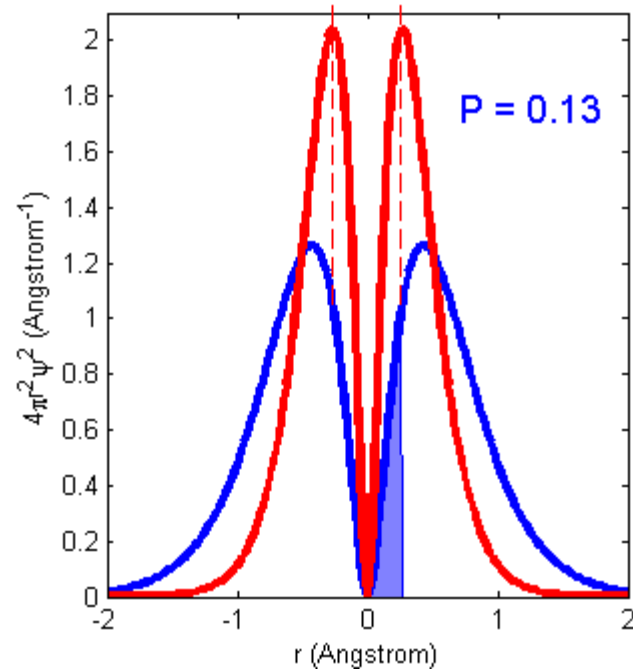
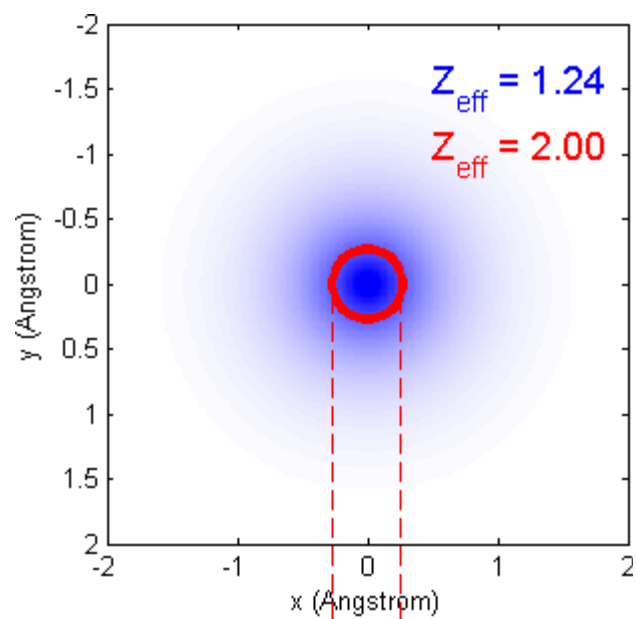
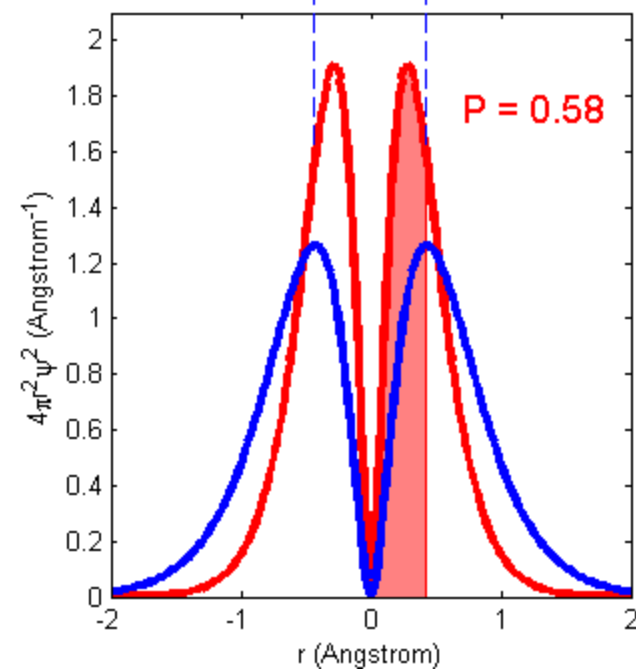
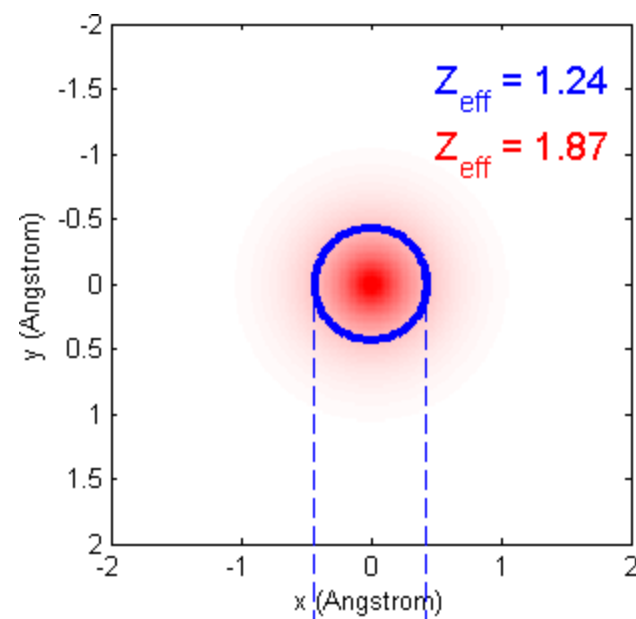
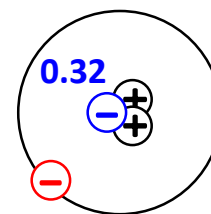
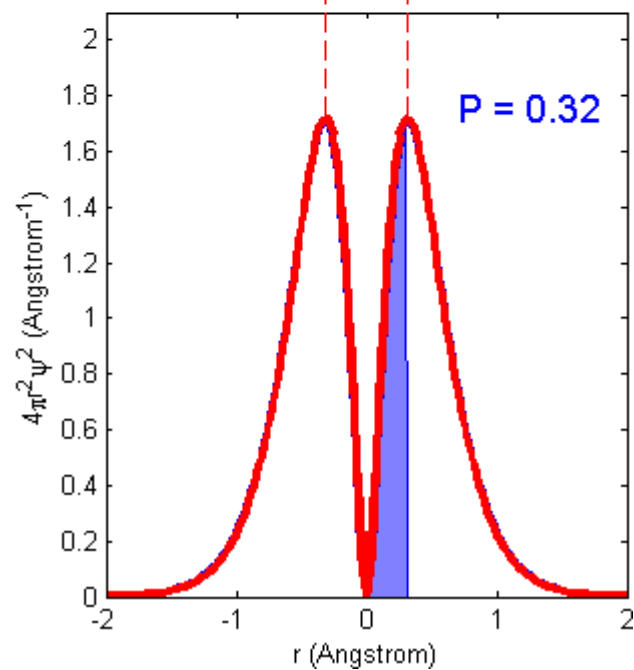
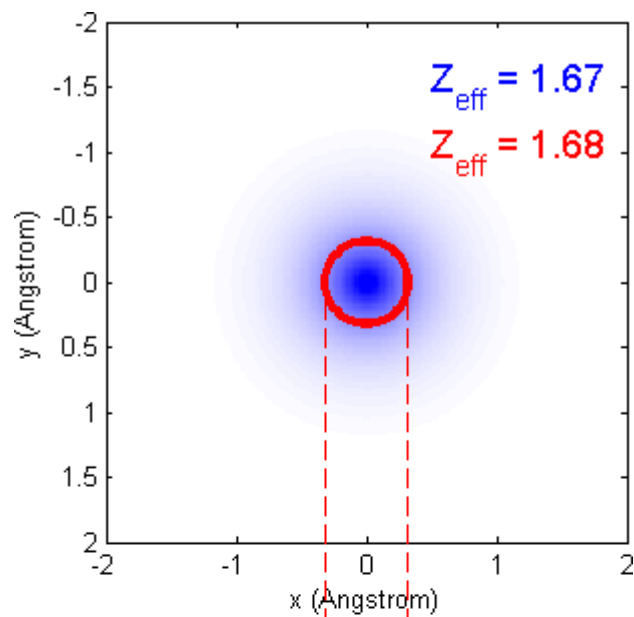


Diagram of a nucleus with a blue circle labeled 0.13 and a red circle labeled $-$. The nucleus is represented by a black circle with a $+$ sign.

$$Z_{\text{eff}} = 2 - 0.13$$

$$= 1.87$$

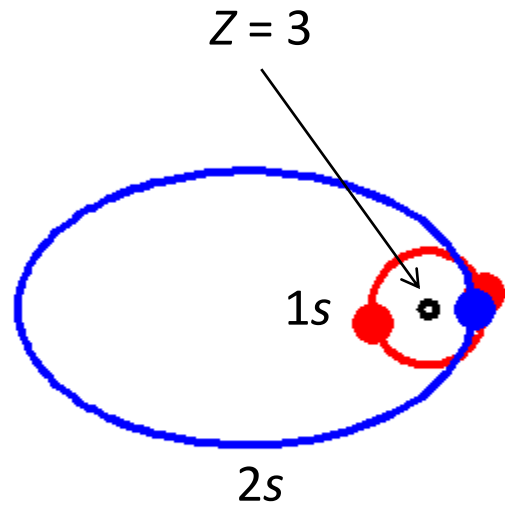




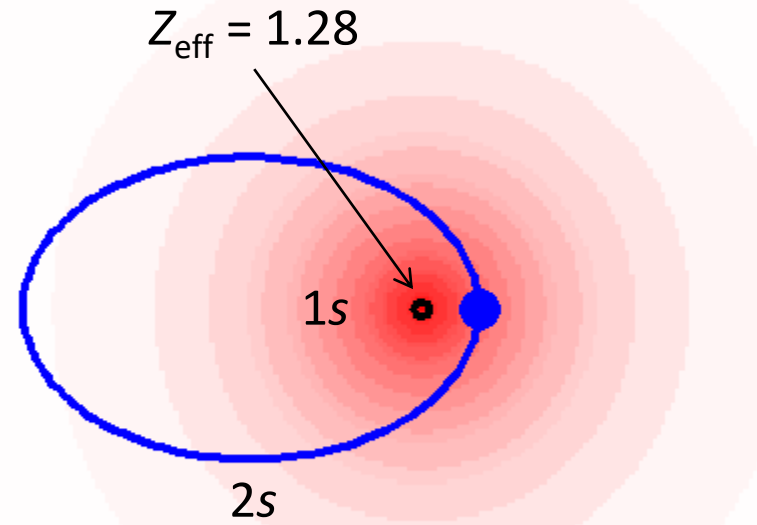
$$Z_{\text{eff}}(\text{He } 1s) = 2 - 0.32 = 1.68$$

Screening effect
(Shielding effect)
屏蔽效应

The Shell Model for Li



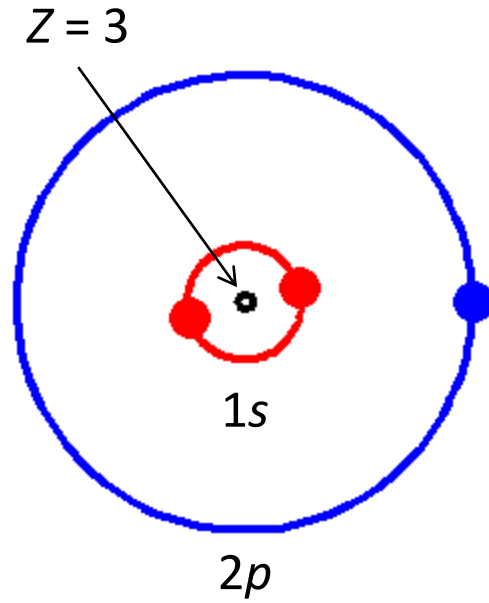
Li: $\psi(r_1, r_2, r_3)$



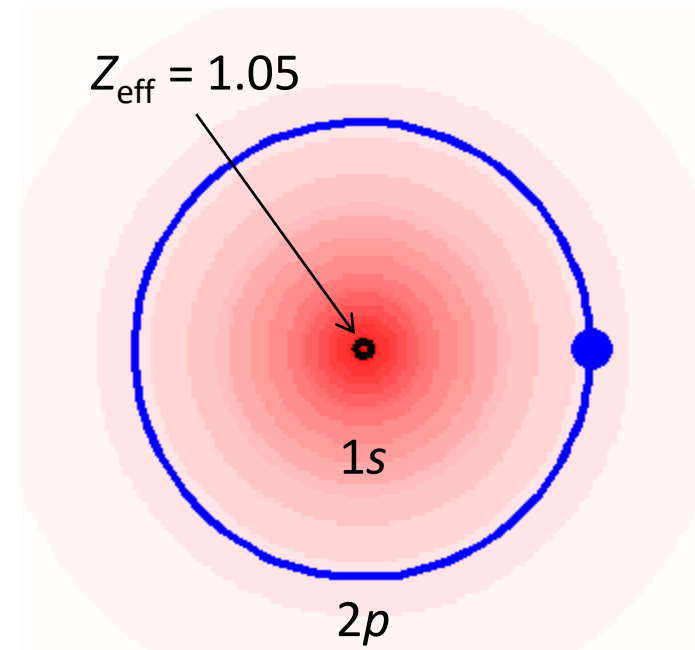
Li: $\varphi_{1s}(r_1)^2 \cdot \varphi_{2s}(r_3)$

$1s^2 2s^1$

The Shell Model for Li



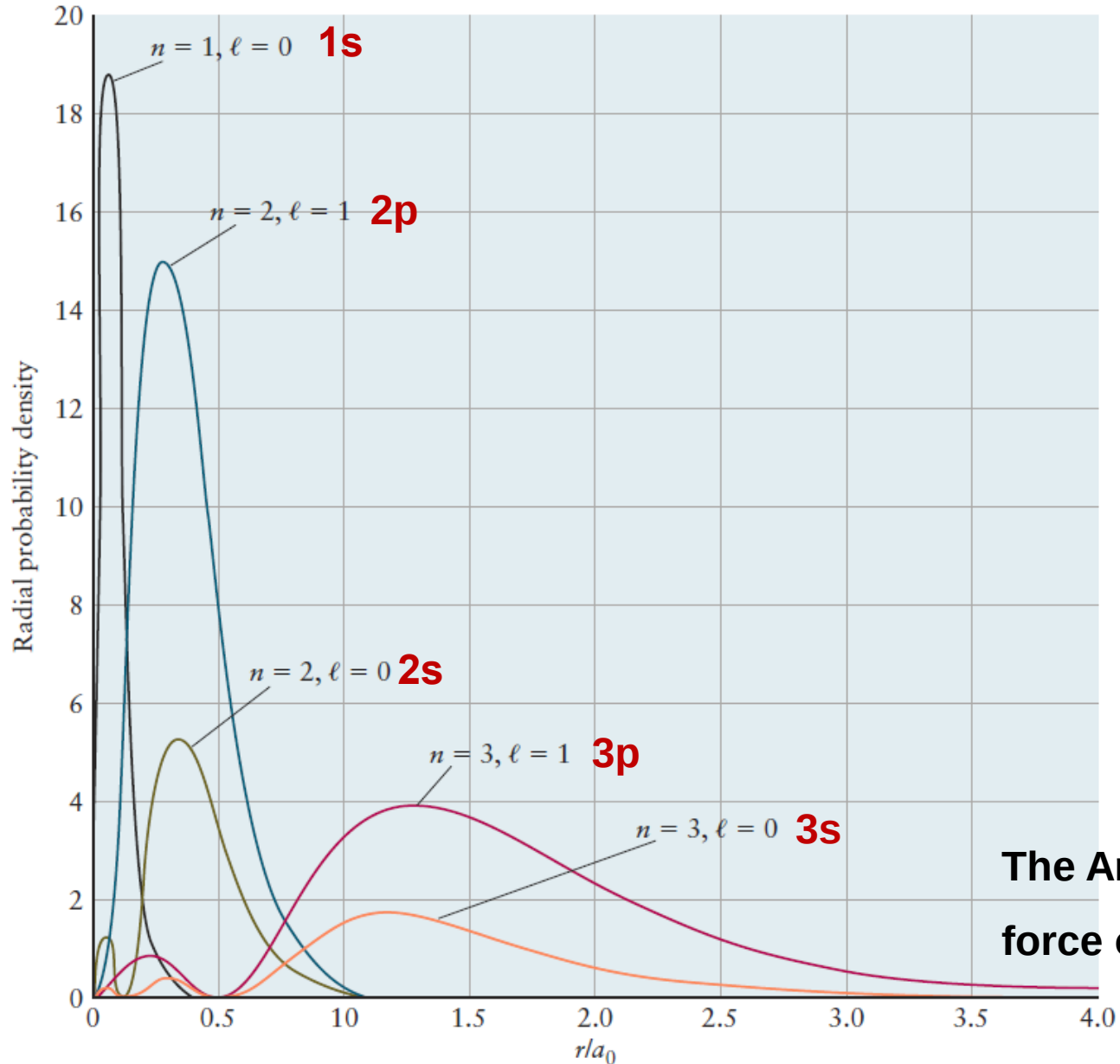
*Li: $\psi(r_1, r_2, r_3)$



*Li: $\varphi_{1s}(r_1)^2 \cdot \varphi_{2p}(r_3)$

$1s^2 2p^1$

Hartree Orbitals: sizes and shapes

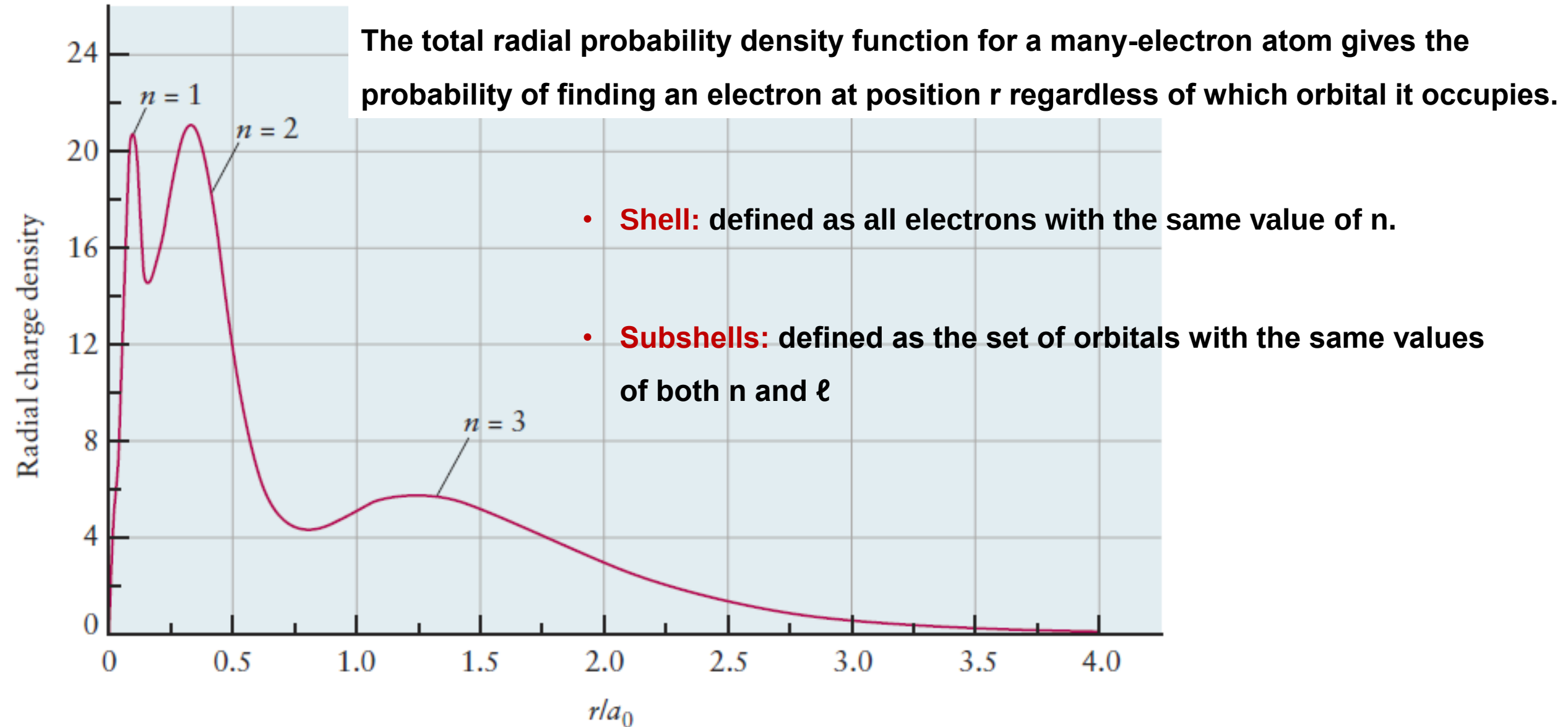


Hartree orbitals in Ar with $n = 1, 2, 3$.

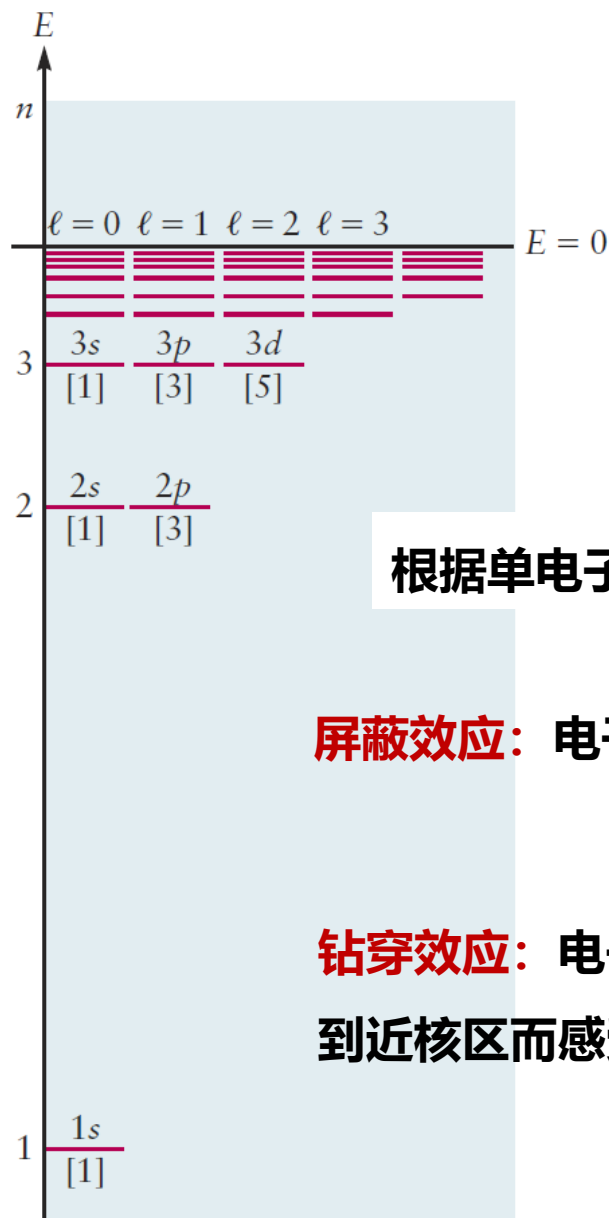
1s; 2s, 2p; 3s, 3p

The Ar nucleus ($Z=18$) exerts a much stronger attractive force on electrons than does the H nucleus ($Z=1$).

Hartree Orbitals: sizes and shapes



Hartree Orbitals: sizes and shapes



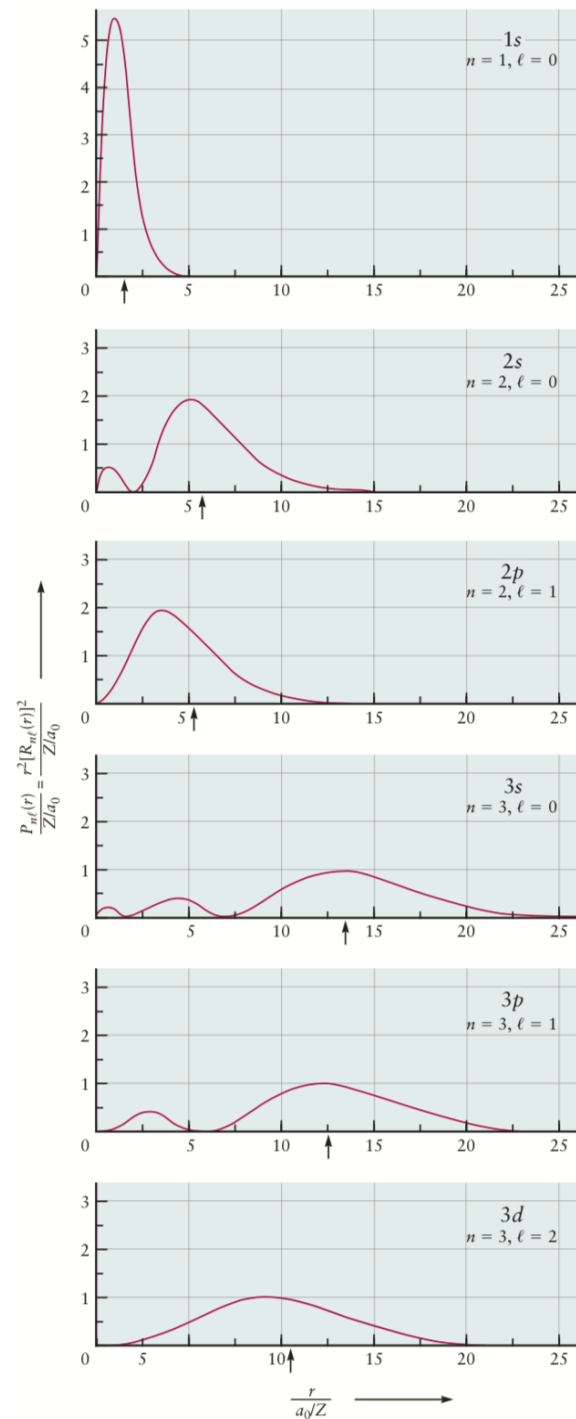
根据单电子波函数和中心力场的近似模型：

屏蔽效应：电子 i 受到核电荷数的减少，能级升高

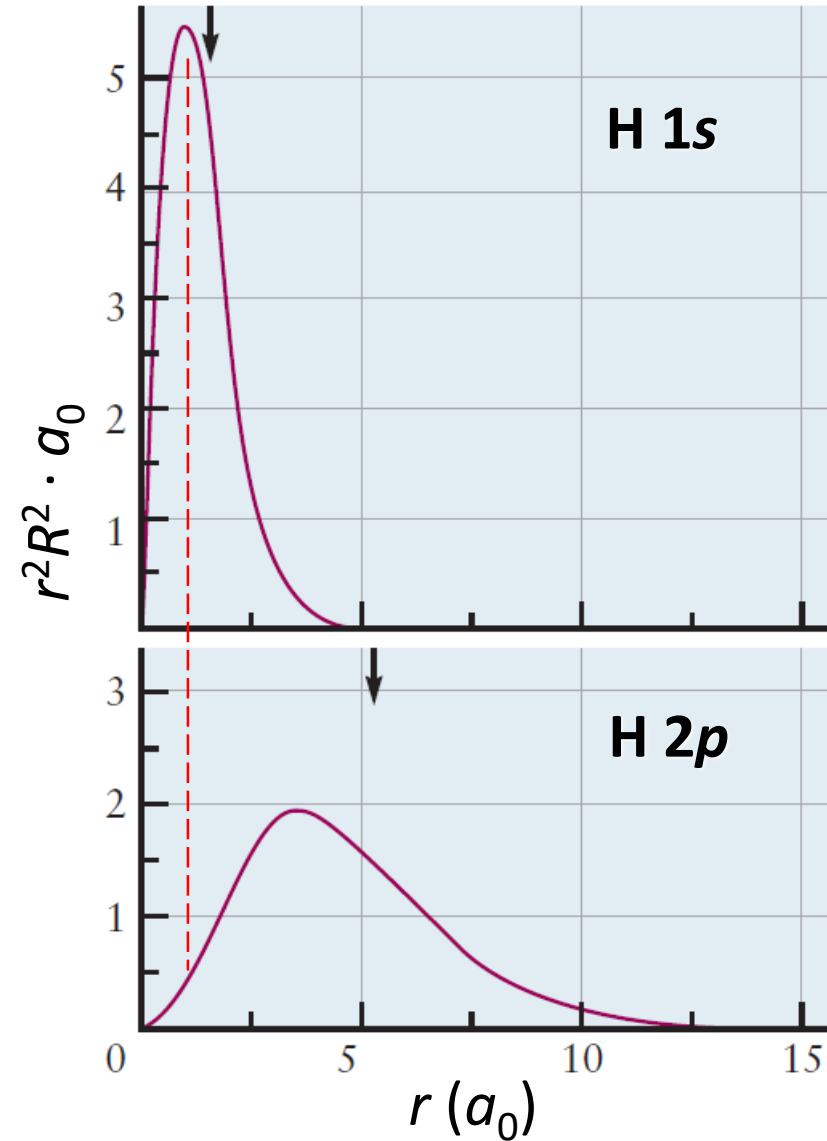
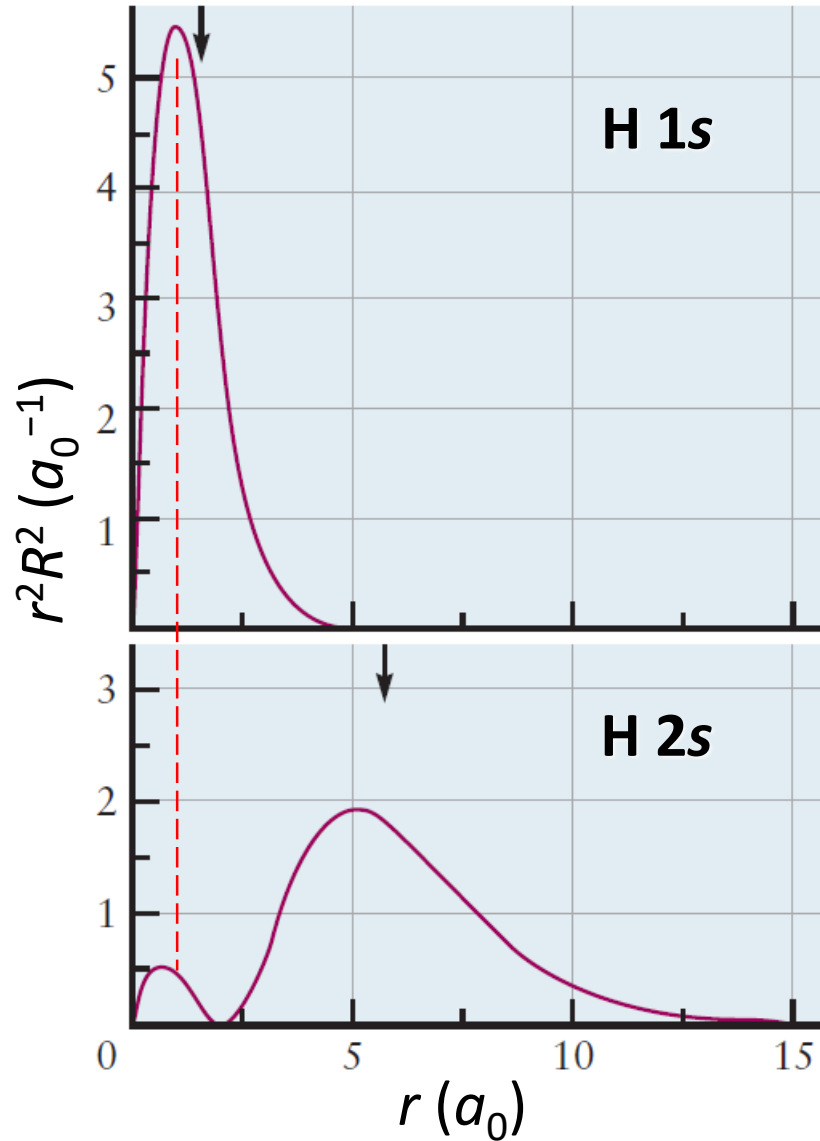
钻穿效应：电子 i 避开其余电子的屏蔽，其电子云钻到近核区而感受到较大的核电荷作用，能级降低

径向分布图

$$\frac{P_{n\ell}(r)}{Z/a_0} = \frac{r^2 [R_{n\ell}(r)]^2}{Z/a_0}$$



The Penetration Effect 钻穿效应



Shielding Effects: Energy Sequence of Hartree Orbitals

The dependence of the energy on ℓ and n can be explained by comparing the extent of shielding in different subshells.

$$\epsilon_{ns} < \epsilon_{np} < \epsilon_{nd}$$

- Only the **s orbitals penetrate** to the nucleus
- Both **p and d orbitals have nodes at the nucleus**
- Consequently, the shielding will be smallest, and the electron most tightly bound, in s orbitals

Shielding Effects: Energy Sequence of Hartree Orbitals

Assume each of the electrons in shell n is moving in a Coulomb potential given approximately by

$$V_n^{\text{eff}}(r) \approx -\frac{Z_{\text{eff}}(n)e^2}{r}$$

We use ε_n to distinguish a Hartree orbital energy from the H atom orbital E_n

$$\varepsilon_n \approx -\frac{[Z_{\text{eff}}(n)]^2}{n^2} \quad (\text{rydbergs})$$

$$\bar{r}_{n\ell} \approx \frac{n^2 a_0}{Z_{\text{eff}}(n)} \left\{ 1 + \frac{1}{2} \left[1 - \frac{\ell(\ell + 1)}{n^2} \right] \right\}$$

Shielding Effects: Energy Sequence of Hartree Orbitals

Slater's **empirical rules** for **estimating the shielding constant** for an electron in an ns or np orbital:

1. Write the electronic configuration for the atom with terms **in the order and grouped**: (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) etc.
2. **Electrons in any group to the right of a (ns, np) group do not shield the (ns, np) electrons** and therefore make no contribution to the shielding constant.
3. **Each of the other electrons in a (ns, np) group contributes 0.35** to the shielding constant for the valence electron.
4. **Each electron in the (n-1) shell contributes 0.85** to the shielding constant.
5. **Each electron in shells (n-2) or lower fully shield the outer electrons**, contributing 1.00 to the shielding constant.

Shielding Effects: Energy Sequence of Hartree Orbitals

Orbital Energies (×13.6 eV)

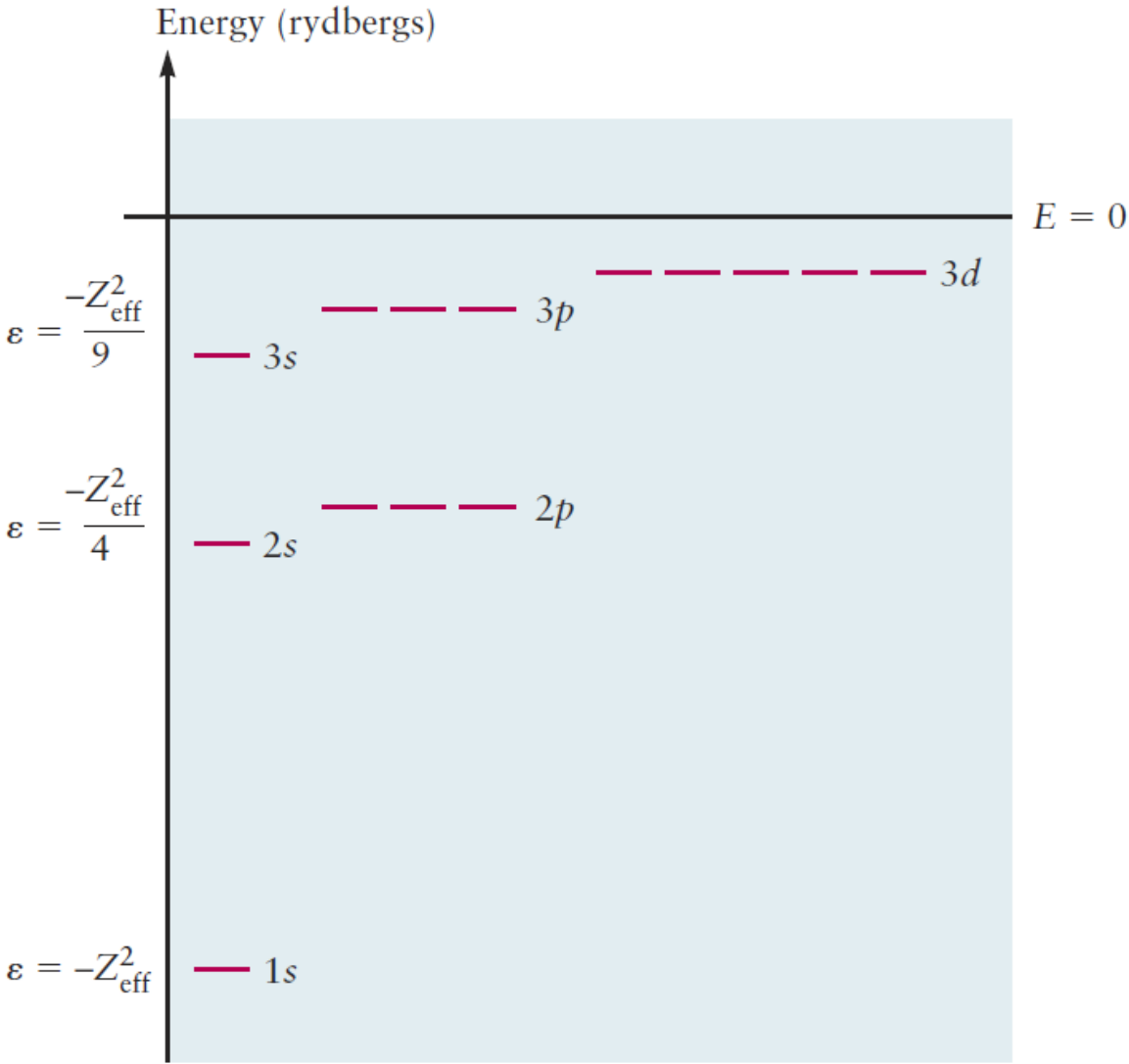
	¹ H	² He ⁺	³ Li ²⁺	⁴ Be ³⁺	⁵ B ⁴⁺
1s	−1	−2 ²	−3 ²	−4 ²	−5 ²
2s, 2p	−1/2 ²	−(2/2) ²	−(3/2) ²	−(4/2) ²	−(5/2) ²

	H	He	Li	Be	B
1s	−1	−1.65 ²	−2.65 ²	−3.65 ²	−4.65 ²
2s			−(1.3/2) ²	−(1.95/2) ²	−(2.6/2) ²
2p					−(2.6/2) ²

TABLE 5.3
Z_{eff} for Selected Atoms

	H(1)							He(2)
1s	1.00							1.69
	Li(3)	Be(4)	B(5)	C(6)	N(7)	O(8)	F(9)	Ne(10)
1s	2.69	3.68	4.68	5.67	6.66	7.66	8.65	9.64
2s	1.28	1.91	2.58	3.22	3.85	4.49	5.13	5.76
2p			2.42	3.14	3.83	4.45	5.10	5.76

$$\epsilon_n \approx -\frac{[Z_{\text{eff}}(n)]^2}{n^2} \quad (\text{rydbergs})$$



Major points of today's lecture

- **Quantum mechanics and atomic structure**
 1. The hydrogen atom
 2. Shell model for many-electron atoms
 3. Aufbau principle and electron configuration
 4. Shells and the periodic table: photoelectron spectroscopy
 5. Periodic properties and electronic structure

Aufbau principle (构造原理) and electron configuration

Li atom

$$\psi = \psi(x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3)$$

The orbital approximation for atoms

$$\psi_{\text{atom}} = \varphi_{\alpha}(r_1)\varphi_{\beta}(r_2)\varphi_{\gamma}(r_3)$$

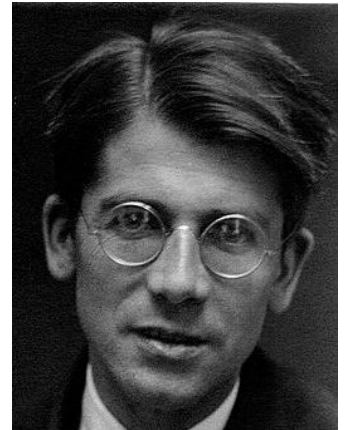
The **ground-state electronic configuration** of an atom (atomic number Z):
arrange the Hartree atomic orbitals in order of increasing energy, until all
Z electrons are in place.

The **Pauli exclusion principle**: no two electrons in an atom can have the same
set of four quantum numbers (n, ℓ , m, m_s).

Hund's rules: when electrons are added to Hartree orbitals of equal energy, a
single electron enters each orbital before a second one enters any orbital. In
addition, the lowest energy configuration is the one **with parallel spins**.



Wolfgang Pauli
(1900–1958)



Friedrich Hund
(1896–1997)



Photo from the Nobel
Foundation archive.

Wolfgang Pauli

Prize share: 1/1

The Nobel Prize in Physics 1945 was awarded to Wolfgang Pauli "for the discovery of the Exclusion Principle, also called the Pauli Principle"

<https://www.nobelprize.org/prizes/physics/1954/summary/>

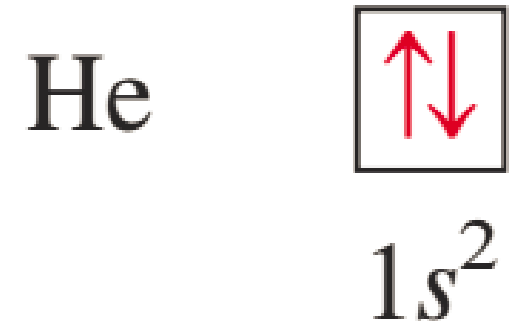
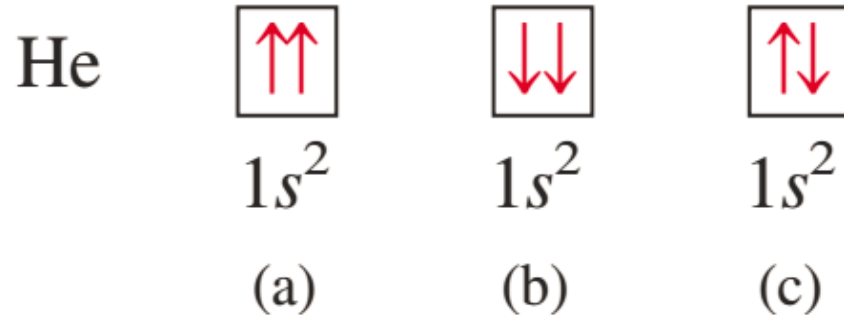
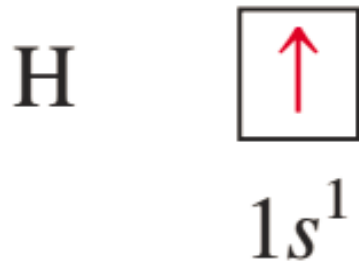
Pauli exclusion principle

denotes the principal quantum number n

denotes the number of electrons in the orbital or subshell

denotes the angular momentum quantum number ℓ

$1s^1$

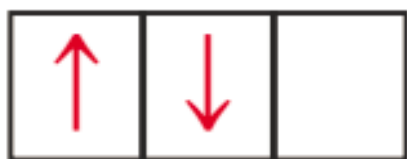


$(1, 0, 0, +\frac{1}{2})$ $(1, 0, 0, -\frac{1}{2})$

Hund's Rule



$2p_x$ $2p_y$ $2p_z$



$2p_x$ $2p_y$ $2p_z$



$2p_x$ $2p_y$ $2p_z$

C



$1s^2$



$2s^2$



$2p^2$

O



$1s^2$



$2s^2$



$2p^4$

F



$1s^2$



$2s^2$



$2p^5$

N



$1s^2$



$2s^2$



$2p^3$

Ne



$1s^2$

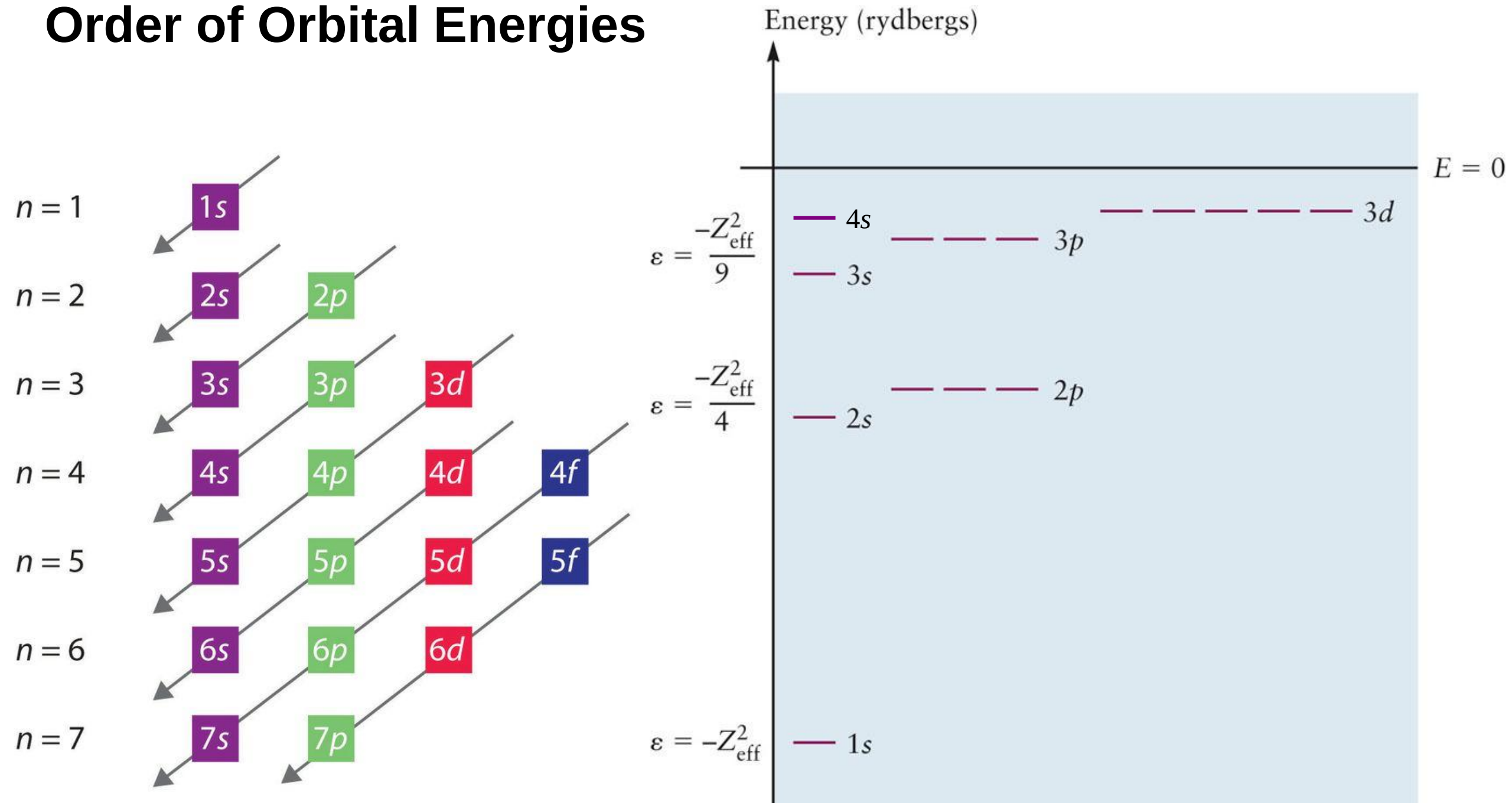


$2s^2$



$2p^6$

Order of Orbital Energies



Electron Configurations of H thru Ar

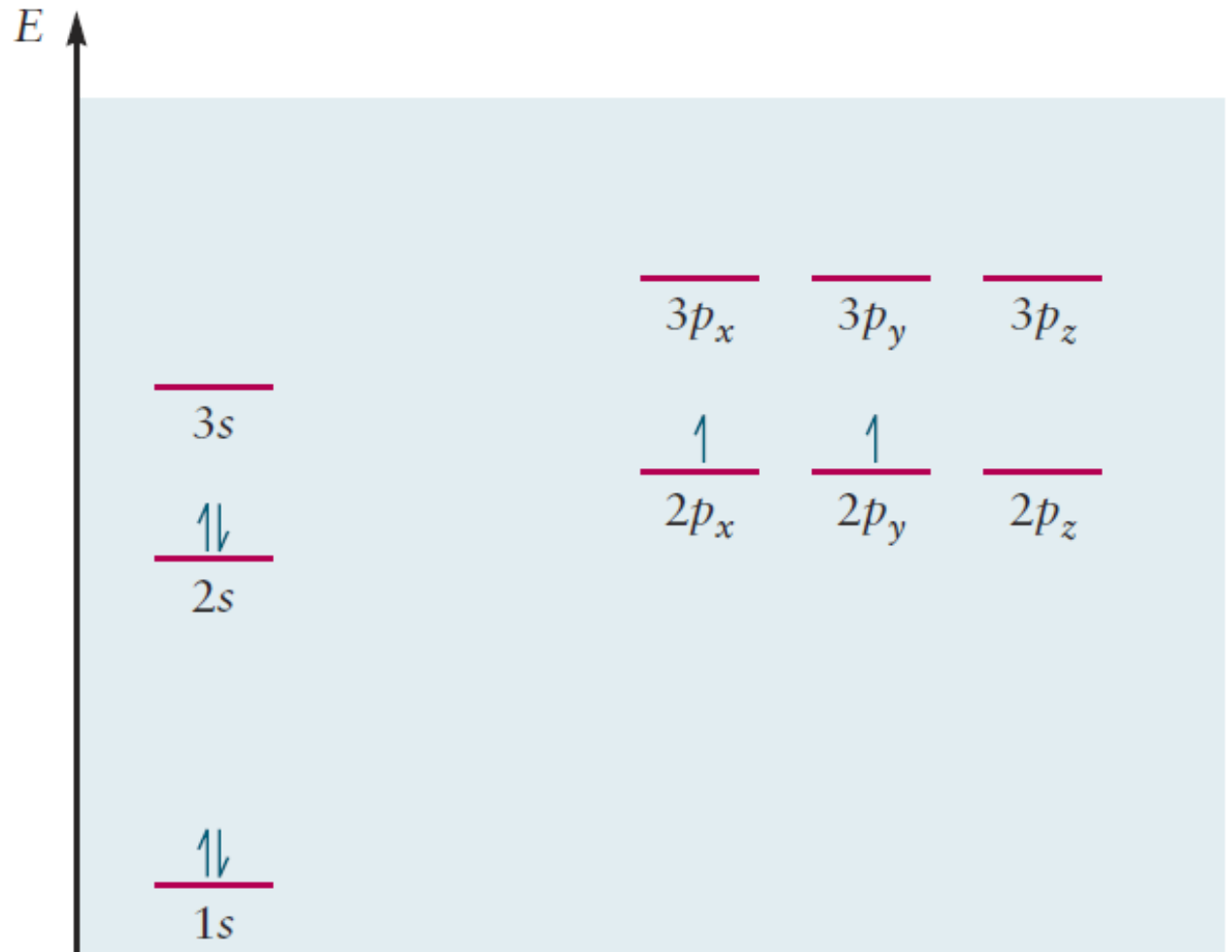
	1s	2s	2p _x	2p _y	2p _z
H: 1s ¹	<u>↑</u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>
He: 1s ²	<u>↑↓</u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>
Li: 1s ² 2s ¹	<u>↑↓</u>	<u>↑</u>	<u> </u>	<u> </u>	<u> </u>
Be: 1s ² 2s ²	<u>↑↓</u>	<u>↑↓</u>	<u> </u>	<u> </u>	<u> </u>

The four elements from hydrogen to beryllium are called **s-block elements**: their configurations involve filling of s orbitals in the building-up process.

Electron Configurations of H thru Ar

Two electrons that occupy the same atomic orbital experience stronger electron-electron repulsion than they would if they occupied orbitals in different regions of space.

FIGURE 5.20 For a many-electron atom such as carbon, orbitals with different ℓ values (such as the $2s$ and $2p$ orbitals) have different energies (see Fig. 5.18). When two or more orbitals have the *same* energy (such as the three $2p$ orbitals here), electrons occupy different orbitals with parallel spins in the ground state.



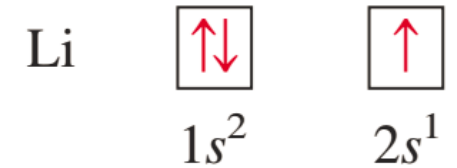
Electron Configurations of H thru Ar

	1s	2s	2p _x	2p _y	2p _z
B: 1s ² 2s ² 2p _x ¹	<u>↑↓</u>	<u>↑↓</u>	<u>↑</u>	<u> </u>	<u> </u>
C: 1s ² 2s ² 2p _x ¹ 2p _y ¹	<u>↑↓</u>	<u>↑↓</u>	<u>↑</u>	<u>↑</u>	<u> </u>
N: 1s ² 2s ² 2p _x ¹ 2p _y ¹ 2p _z ¹	<u>↑↓</u>	<u>↑↓</u>	<u>↑</u>	<u>↑</u>	<u>↑</u>
O: 1s ² 2s ² 2p _x ² 2p _y ¹ 2p _z ¹	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>	<u>↑</u>	<u>↑</u>
F: 1s ² 2s ² 2p _x ² 2p _y ² 2p _z ¹	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>	<u>↑</u>
Ne: 1s ² 2s ² 2p _x ² 2p _y ² 2p _z ²	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>	<u>↑↓</u>

The six elements from boron to neon are called **p-block elements**:
 their configurations involve filling of p orbitals in the building-up process.

Electron Configurations of H thru Ar

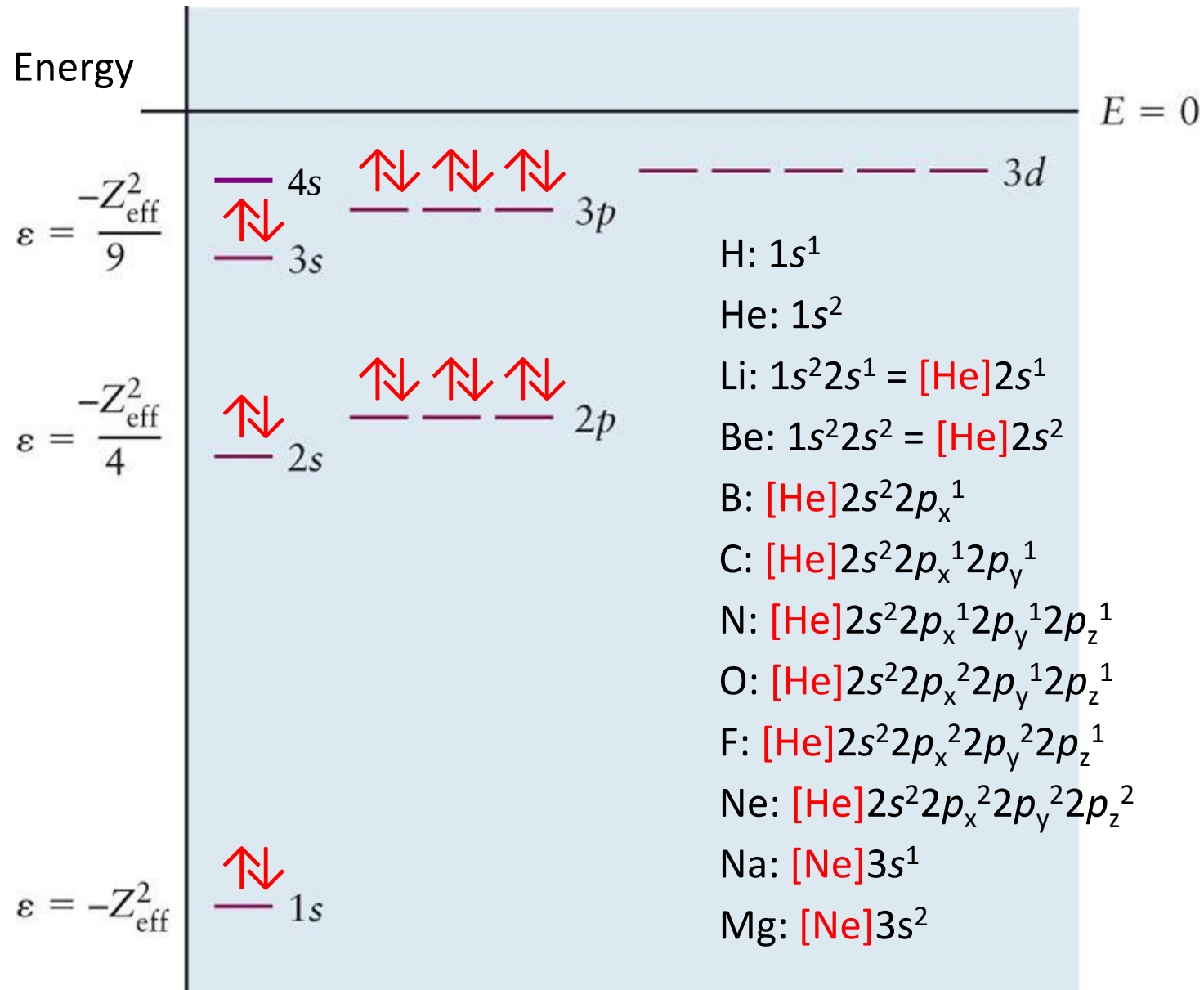
A substance is **paramagnetic (顺磁性)** if it is **attracted into a magnetic field**. Any substance that has **one or more unpaired electrons** in the atoms, molecules, or ions that compose it is paramagnetic because a net magnetic moment arises from each of the unpaired electrons.



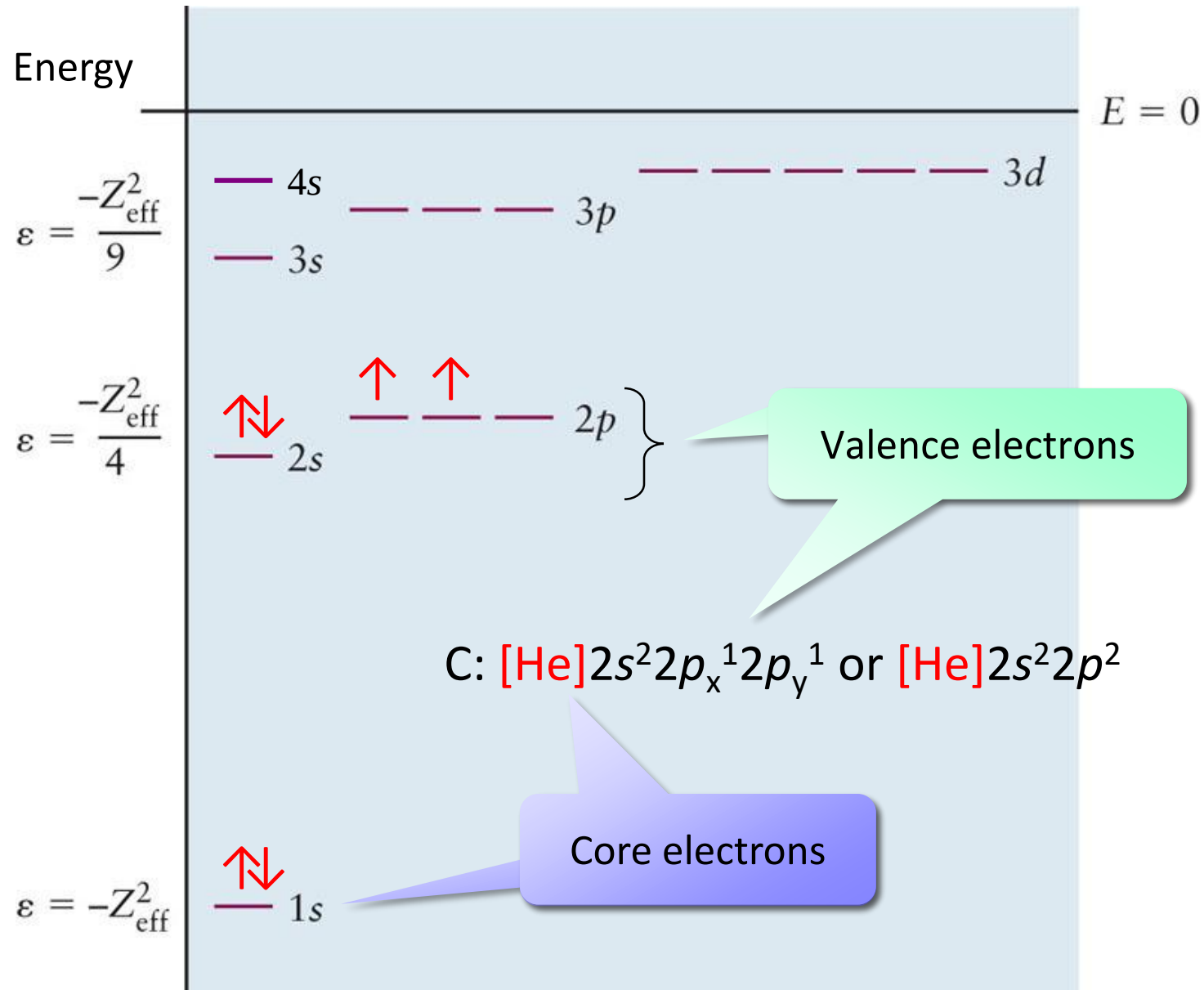
A substance in which all the electrons are paired is weakly **diamagnetic (反磁性)**. It is **pushed out of a magnetic field**, although the force it experiences is much smaller in magnitude than the force that pulls a typical paramagnetic substance into a magnetic field.



Electron Configurations of H thru Ar



Core and Valence Electrons



Write the ground-state electron configurations for magnesium and sulfur. Are the gaseous atoms of these elements paramagnetic or diamagnetic?

The boron atom with $Z = 5$ has electron configuration $\text{B}: (1s)^2(2s)^2(2p_x)^1$.

- (a) Write the atomic wave function for a B atom.
- (b) Estimate the energy level diagram for a B atom.
- (c) Estimate the radius of the $2s$ and $2p_x$ orbitals.

Transition-Metal Elements and Beyond

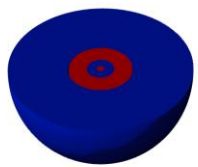
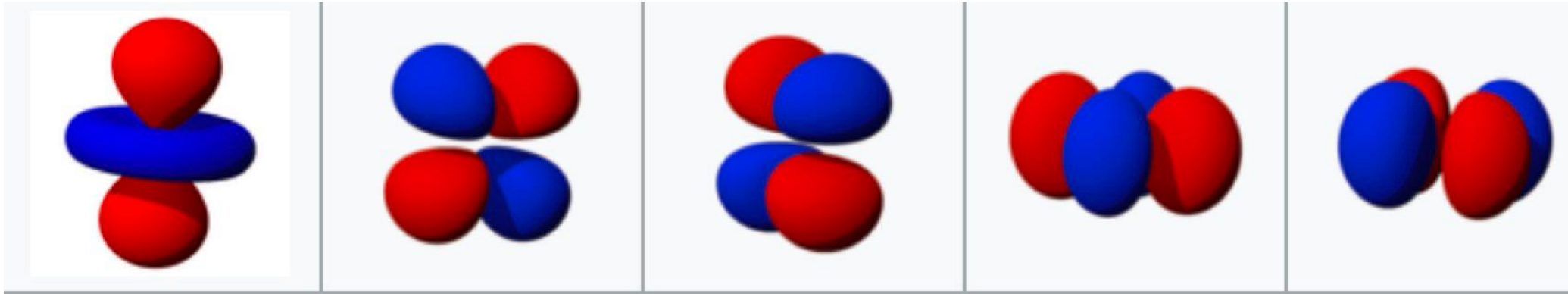
Advanced calculations for elements 19 (K) through 30 (Zn) predict that ϵ_{3d} and ϵ_{4s} are very close, so the build-up process becomes rather subtle.

Optical spectroscopy confirms that the ground state K is $[\text{Ar}]3d^04s^1$ and that of Ca is $[\text{Ar}]3d^04s^2$, as predicted by the sequence of calculated orbital energies $\epsilon_{3d} > \epsilon_{4s}$

For Sc and the elements beyond, advanced calculations predict that $\epsilon_{3d} < \epsilon_{4s}$

Filling the 3d orbitals first would give the configurations $[\text{Ar}]3d^34s^0$ for Sc, $[\text{Ar}]3d^44s^0$ for Ti, and so on to $[\text{Ar}]3d^{10}4s^0$ for Ni according to calculation. However, experiments suggest $\epsilon_{3d} > \epsilon_{4s}$

Transition-Metal Elements and Beyond



Because the 3d orbital is much more localized than the 4s orbital, the much greater repulsion energy of the two electrons in the 3d orbital outweighs the fact that $\epsilon_{3d} < \epsilon_{4s}$.

For Sc: **[Ar]3d¹4s²** vs. [Ar]3d³4s⁰

电子互斥能

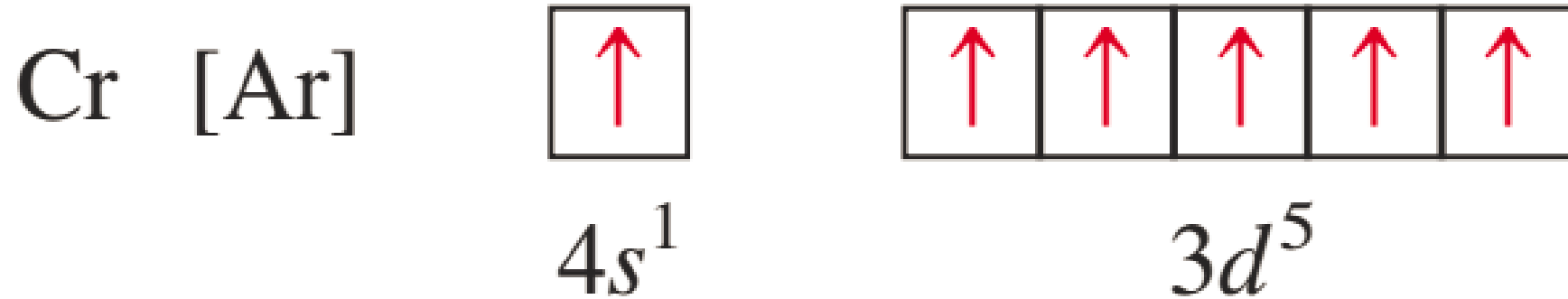
$$J(d,d) = 11.78\text{eV}$$

$$J(d,s) = 8.38\text{eV}$$

$$J(s,s) = 6.60\text{eV}$$

The ten elements from scandium to zinc are called **d-block elements**: their configurations involve the filling of a d orbital in the building-up process.

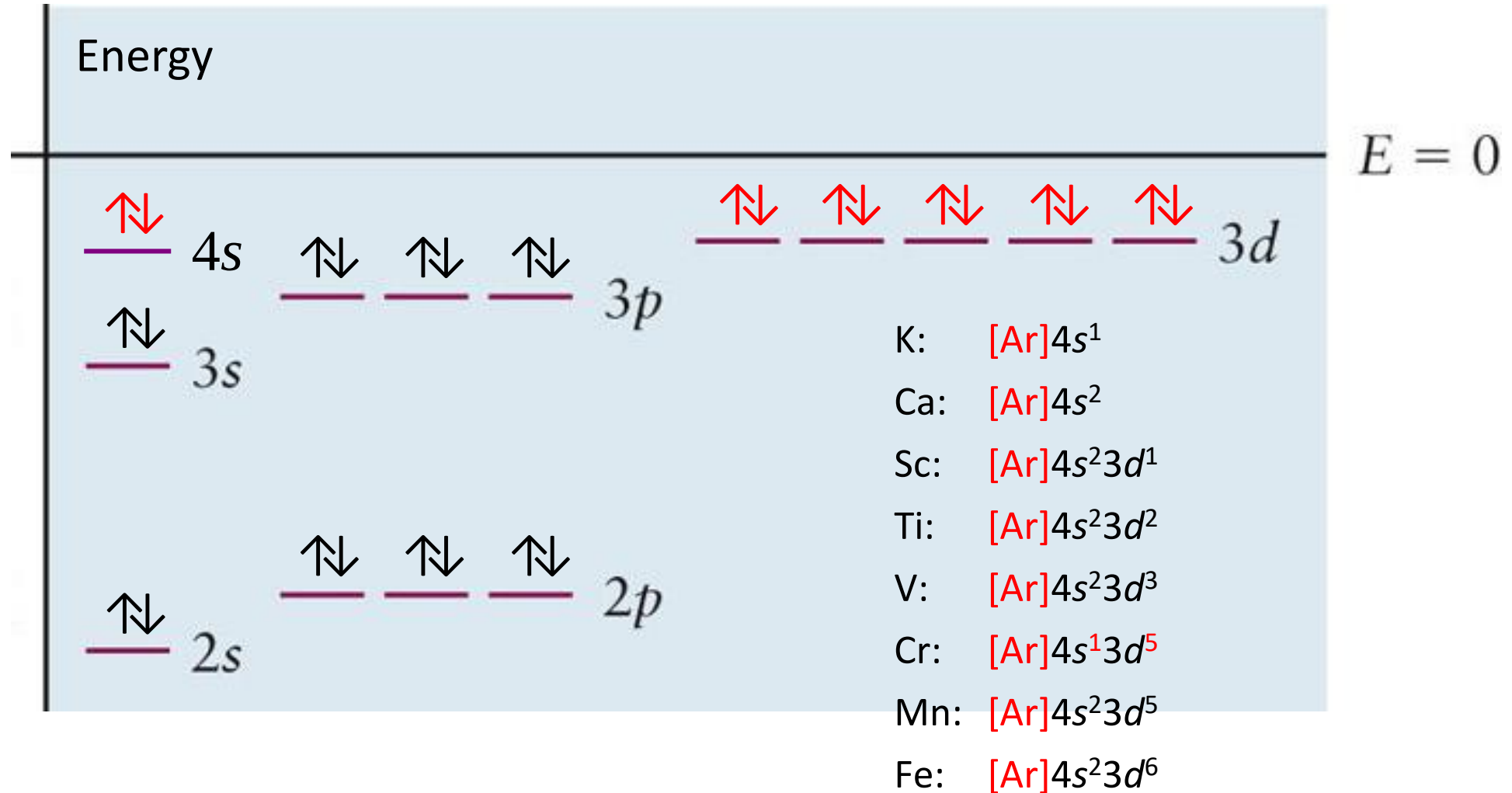
Electron Configurations of K thru Zn



Two exceptions



Electron Configurations of K thru Zn



Classification of groups of elements

1s																		1s	
H																		He	
2s–filling												2p–filling							
Li	Be											B	C	N	O	F	Ne		
3s–filling												3p–filling							
Na	Mg											Al	Si	P	S	Cl	Ar		
4s–filling		3d–filling										4p–filling							
K	Ca	Sc	Ti	V	Cr 3d⁵4s¹	Mn	Fe	Co	Ni	Cu 3d¹⁰4s¹	Zn	Ga	Ge	As	Se	Br	Kr		
5s–filling		4d–filling										5p–filling							
Rb	Sr	Y	Zr	Nb 4d⁴5s¹	Mo 4d⁵5s¹	Tc	Ru 4d⁷5s¹	Rh 4d⁸5s¹	Pd 4d¹⁰	Ag 4d¹⁰5s¹	Cd	In	Sn	Sb	Te	I	Xe		
6s–filling		5d–filling										6p–filling							
Cs	Ba	Lu	Hf	Ta	W	Re	Os	Ir	Pt 5d⁹6s¹	Au 5d¹⁰6s¹	Hg	Tl	Pb	Bi	Po	At	Rn		
7s–filling		6d–filling																	
Fr	Ra	Lr	Rf	Ha	Sg	Ns	Hs	Mt	Uun	Uuu									

4f–filling													
La 5d¹6s²	Ce 4f¹5d¹6s²	Pr	Nd	Pm	Sm	Eu	Gd 4f⁷5d¹6s²	Tb	Dy	Ho	Er	Tm	Yb
5f–filling													
Ac 6d¹7s²	Th 6d²7s²	Pa 5f²6d¹7s²	U 5f³6d¹7s²	Np 5f⁴6d¹7s²	Pu	Am	Cm 5f⁷6d¹7s²	Bk	Cf	Es	Fm	Md	No

Transition-Metal Elements and Beyond

In the sixth period, the filling of the 4f orbitals (and the generation of **the f-block elements**) begins as the rare-earth (lanthanide) elements from lanthanum to ytterbium are reached.

The configurations determined from calculations and experiment can be recalled as needed by assuming that the orbitals are filled in the sequence:

$1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s \rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d$.

The energies of the 4f, 5d, and 6s orbitals are **comparable over much of the sixth period**, and thus their order of filling is erratic.

Homework: read “principle of modern chemistry”, chapter 5.2; 5.3